

# An Exponential Lower Bound for Exact Parity in Depth-Four QAC Circuits

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## Abstract

We prove an exponential lower bound for exact computation of parity by depth-four QAC circuits. The model permits arbitrary one-qubit gates, arbitrary-arity generalized Toffoli gates, an arbitrary finite number of product-initialized ancillas, and unrestricted garbage outside a single output qubit. The proof reduces exact computation to exponential decay of full-parity coefficients for product-measurement events after two layers. The principal ingredients are a constant mixedness budget for a one-input first-layer component, a rank-one insertion inequality, an exact expansion of high-weight second-layer gates, and a bounded-differences estimate that does not assume independence between different input components. We also prove that a bounded-error depth-four family with fixed positive advantage must have quadratic arity at the final gate incident to the output. The argument does not rule out bounded-error depth four with arbitrary polynomial final arity.

## 1 Introduction

### 1.1 Motivation, model, and main result

Define  $\text{PARITY}_n(x) = x_1 \oplus \cdots \oplus x_n$  and  $\chi(x) = (-1)^{\text{PARITY}_n(x)}$  for  $x \in \{0, 1\}^n$ . Parity tests the ability of a shallow circuit to combine information from all its inputs. For each fixed classical unbounded-fan-in AND/OR depth  $D \geq 2$ , with input negations free, Håstad's switching-lemma method gives a size lower bound  $\exp(\Omega(n^{1/(D-1)}))$  for computing parity [1]. This is stretched exponential when  $D > 2$ , as distinct from the  $\exp(\Omega(n))$  bound proved here. The low-degree Fourier concentration theorem of Linial, Mansour, and Nisan [2] gives a complementary analytic description of the classical limitation. A quantum counterpart must accommodate interference and entanglement, not just classical restrictions.

Moore [3] introduced  $\text{QAC}^0$  using arbitrary one-qubit gates and unbounded-arity generalized Toffoli gates at constant depth, with polynomially many gates and ancillary wires; see also Green, Homer, Moore, and Pollett [4] and the survey of Bera, Green, and Homer [5]. The connection with fan-out makes parity particularly significant: the coherent parity and fan-out unitaries are related by local Hadamard conjugation, and their implementation is connected to cat-state preparation [3, 4]. Fan-out is an additional resource, not a primitive of the model considered here. Høyer and Špalek [6] demonstrated its power for constant-depth approximate computation, while Takahashi and Tani [7] proved the exact fan-out-assisted class collapse  $\text{QNC}_f^0 = \text{QAC}_f^0 = \text{QTC}_f^0$ . The subscript  $f$  means

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that unbounded fan-out is supplied as a primitive; the three classes use bounded-arity, generalized-Toffoli, and threshold gates, respectively, in addition to one-qubit gates. These results motivate understanding what remains possible without unbounded fan-out.

Two distinctions are essential. First, an unbounded-arity gate can meet all input wires in one layer, so counting wires in an ordinary bounded-fan-in backward light cone cannot itself give a lower bound. Second, producing the Boolean parity bit on one output qubit, with arbitrary garbage elsewhere, is a less restrictive task than implementing the coherent parity unitary while preserving the input registers. Our lower bounds address the former task.

The main result, Theorem 1, is that exact single-output parity on  $n$  inputs at entangling depth at most four requires  $M \geq e^{c_* n}$  multiqubit gates for an absolute  $c_* > 0$  and all sufficiently large  $n$ . Here QAC denotes the gate-set model without a polynomial resource restriction, whereas QAC<sup>0</sup> denotes polynomial-size constant-depth families. There is no numerical restriction on gate arities or on the finite number of ancillary wires, initialized in fixed pure product states independent of the input. No bounded incidence degree or treewidth is assumed. Inputs need not be preserved, ancillas need not be uncomputed, and the output may itself be an original input wire. Depth counts disjoint multiqubit-gate layers; intervening one-qubit unitaries are free. Size counts multiqubit gates, so the bound also applies when one-qubit gates are counted. Section 2 states these conventions precisely. The main issue is therefore not how many wires a gate reaches, but how much parity information survives the successive layers.

## 1.2 Related work and comparison

**Ancilla-restricted lower bounds at general depths.** Fang, Fenner, Green, Homer, and Zhang [8] proved logarithmic depth lower bounds for clean exact implementations of parity and fan-out with constantly many ancillas, together with a depth–ancilla tradeoff. Here clean means that the ancillary registers are returned to their initial zero state. Their convention counts one-qubit gates in the depth; the logarithmic conclusion is unchanged by the constant-factor conversion to our convention. Bera [9] gave a different lower-bound method excluding ancilla-free constant-depth exact parity circuits.

Nadimpalli, Parham, Vasconcelos, and Yuen [10] developed Pauli-spectrum concentration, in the spirit of the classical Fourier-analytic approach. For polynomial-size circuits of fixed depth  $d$ , their Theorem 33 implies uniform-input parity success at most  $1/2 + 2^{-\Omega(n^{1/d})}$  when the ancilla count is a sufficiently small constant multiple of  $n^{1/d}$ . Anshu, Dong, Ou, and Yao [11] used spectral-norm approximation by low-degree operators to obtain the stated ancilla threshold  $\tilde{\Omega}(n^{1+3^{-d}})$  for parity. Dong, Ou, and Yao [12] improved the corresponding parity threshold to  $\tilde{\Omega}(n^{1+2^{-d}})$  using operator and unitary dilation; in the small-ancilla regime of their Corollary 3.3, uniform-input success is at most  $1/2 + O(d/n)$ . The tilde suppresses logarithmic factors in these resource thresholds. At depth four, the two stated superlinear thresholds have exponents  $82/81$  and  $17/16$ . These methods apply across arbitrary constant depths and allow arbitrary input-independent ancillary states. Our absence of an ancilla-count restriction must instead be read together with product initialization.

The quoted exponents are the papers’ simplified corollary thresholds, not optimal inversions of their estimates. At fixed depth, the cited estimates give a polynomial that uniformly approximates the single-output acceptance probability and has degree  $\tilde{O}(N^{1-\alpha})$ , where  $N$  is the total number of wires and  $\alpha = 3^{-d}$  or  $2^{-d}$ , respectively. For any fixed positive success advantage, they therefore imply an ancillary requirement  $\tilde{\Omega}(n^{1/(1-\alpha)})$ ; see [11, Theorem 4.1] and [12, Theorem 3.1]. Indeed, the parity character  $\chi$  is orthogonal on the uniform cube to every polynomial of degree less than  $n$ ; approximating an acceptance function uniformly to error smaller than its success advantage therefore forces the degree bound to reach  $n$ . Inverting that bound, and subtracting the  $n$  input

wires, gives the stated requirement. Thus the directly derived depth-four exponents are  $81/80$  and  $16/15$ , respectively.

**Fixed-depth lower bounds and approximate constructions.** Rosenthal [13, Corollary 1.6 and Theorem 1.7] studied the coherent, input-preserving parity operation, allowing final garbage. He obtained an  $\Omega(n/(d+1))$  multiqubit-gate requirement for worst-case coherent approximation with squared-overlap fidelity exceeding  $1/2$  by a fixed positive amount. At depth two, fidelity on a specified product quantum input is at most  $1/2 + \exp(-\Omega(n))$ , independently of size. These are coherent fidelity statements, not average-case Boolean-success bounds; allowing garbage does not remove input preservation. His Corollary 1.2 also gives, for every  $d \geq 7$  and  $0 < \varepsilon < 1$ , circuits of size  $\exp(\text{poly}(n^{1/d}) \log(n/\varepsilon))$  whose squared Euclidean error relative to the ideal parity output with zero ancillas is at most  $\varepsilon$  on every pure quantum input. This is his phase-dependent fidelity guarantee, not a Boolean failure probability. Such greater-depth approximate constructions do not contradict an exact depth-four lower bound.

Fenner, Grier, Padé, and Thierauf [14] proved that exact single-output parity on more than three non-target input bits is impossible at depth two, regardless of gate count or the number of product-initialized ancillas. Their proof uses irreducible polynomials and entanglement. Joshi, Tal, Vasconcelos, and Wright [15] proved exact depth-three impossibility for  $n > 100$ , again independently of size and the number of product-initialized ancillas, and a depth-two single-output parity-correlation bound  $\exp(-\Omega(\sqrt{n}))$ . Their exact analysis uses quantum restrictions and classical  $\text{AC}^0$  simulation.

Our exact result increases the lower-bound depth from three to four, but changes its resource conclusion: it requires exponentially many gates, rather than excluding circuits of every size. Compared with general-depth ancilla bounds, it removes any numerical limit on product-initialized ancillary space at this fixed depth. Compared with Rosenthal’s general-depth gate bound, it gives an exponential lower bound for exact computation and applies to the more permissive single-output task. It neither excludes exponentially large depth-four circuits nor establishes a lower bound at every constant depth.

### 1.3 What is novel and what is derived

We distinguish the new technical estimates from their circuit-complexity consequences and from the previously established tools used to prove them. Here a *derived contribution* means a consequence proved from our estimates, not a result already established in the cited literature.

**New technical contributions.** The proof has three principal quantitative ingredients. Here a *source* is a first-layer component with one variable input, including an untouched singleton input. Its *ports* are its intersections with second-layer gates, together with one port for its remaining idle wires.

- (i) *A dimension-independent source budget.* Lemma 2 assigns mixedness weights to the marginals of a one-input first-layer component and bounds their sum by  $C_s = 1/2 + 2/e$ , independently of its number of wires. It also gives a majorant for the squared trace norm of the retained signed source operator (half the difference of the two source states): when each port is independently retained with probability  $1/2$ , this majorant has mean at most  $3/4$ . Changing one port’s retention changes it by at most four times that port’s weight. These estimates make concentration possible even when different sources share the same random choices.
- (ii) *An insertion-stable exponential event estimate.* Lemmas 8 and 9 control joint rank-one insertions, including non-Hermitian correction operators. Combined with an exact expansion of heavy second-layer gates, they give the bound  $K_0 e^{-ar}$ , with absolute  $K_0 \geq 1$  and  $a > 0$ ,

in Theorem 10 for the full-parity coefficient of a product event on  $r$  one-input sources. The constants do not depend on gate count, port dimensions, or the dimension of the independent input-free resource. The theorem even permits that resource to be a non-Hermitian operator of trace norm at most one, as required by subsequent gate expansions. The event may have positive-contraction factors entangled within a source-second-layer port.

- (iii) *A gate-count certificate for exact computation.* Lemma 14 groups final controls by their incidence with the preceding layers. The number of groups is polynomial in the numbers of input sources and gates, even when there are arbitrarily many ancillary wires. This replaces the arity-dependent certificate of Lemma 13 by a gate-count inequality and is the step that removes any ancillary-space bound from the exact theorem.

The novelty claimed here is in these circuit-specific bounds and their combination, not in the elementary reflection expansion, the use of trace norms, or the abstract concentration inequalities.

**Derived contributions and their scope.** The exact exponential lower bound in Theorem 1 follows from the source estimate, the cleanup lemma, and the cell certificate. An additional expansion of large first-layer gates yields Theorem 15, which removes the one-input-per-source condition and gives exponential decay for *physical-qubit* product events. This event class is narrower than the port-product class of Theorem 10; the two hypotheses are not interchangeable.

For a circuit  $C$  with one-qubit output state  $\rho_x$  and a binary measurement represented by a one-qubit Hermitian contraction  $A_o$ , put

$$c(C, A_o) = \mathbb{E}_{x \in \{0,1\}^n} \chi(x) \text{Tr}(\rho_x A_o).$$

The expectation is uniform. Label the binary measurement outcomes by  $\pm 1$ , with  $+1$  denoting even parity. Thus uniform-input success is exactly  $p_{\text{succ}} = (1 + c(C, A_o))/2$ ; see (3) for the full convention.

Pulling back a tensor product of one-qubit contractions supported on  $s$  qubits through a third layer bounds its full-parity coefficient by  $36^s K_0 e^{-\kappa n}$ , for an absolute  $\kappa > 0$ , in Corollary 16. In particular, taking  $s = 1$  gives a size-independent bounded-error depth-three consequence:

$$|c(C, A_o)| \leq 36 K_0 e^{-\kappa n}, \quad p_{\text{succ}} \leq \frac{1}{2} + 18 K_0 e^{-\kappa n}.$$

This conclusion allows arbitrarily many finite gates and product-initialized ancillas. It extends the exact depth-three comparison above to bounded error and strengthens the cited single-output depth-two correlation rate.

At depth four, Corollary 17 gives a different necessary condition. Let  $b$  be the number of other wires in the fourth-layer gate containing the output, and set  $b = 0$  if no such gate is present. In the reflection normal form these wires are controls for a reflection on the output. For  $b \geq 2$ , choosing  $\ell = \lfloor \kappa n / (2 \log 540) \rfloor$  gives the uniform estimate

$$|c(C, A_o)| \leq 108 K_0 e^{-\kappa n/2} + 4e^{\sqrt{2}} \exp\left(-\frac{\kappa n}{\sqrt{b} \log 540}\right).$$

Here and in the proof,  $\log$  denotes the natural logarithm. The cases  $b = 0, 1$  also have exponential decay in  $n$ . Thus bounded  $b$  gives exponential decay,  $b = \Theta(n)$  gives  $\exp(-\Omega(\sqrt{n}))$  decay, and  $b = o(n^2)$  gives vanishing correlation. Remark 18 shows that any fixed uniform-input success advantage  $0 < \delta \leq 1/2$  forces  $b = \Omega_\delta(n^2)$ ; a worst-case success guarantee implies the same condition. Since at most  $n$  controls can be original input wires, this also forces  $\Omega_\delta(n^2)$  ancillary wires. Within the product-initialized depth-four model, this strengthens both the reported  $\tilde{\Omega}(n^{17/16})$  threshold and the  $\tilde{\Omega}(n^{16/15})$  consequence derived above from [12], while locating the required width at the

final output gate. The ancillary-state assumptions differ, so this is not a domination of that result in its full generality.

Quadratic and larger polynomial final arities are not excluded. Consequently the bounded-error conclusion is not an unrestricted depth-four lower bound. The approximate-to-exact reduction of [15, Theorem 1.6] changes depth from  $d$  to  $O(d)$  and does not remove this distinction at depth four.

**Inherited ingredients and attribution.** The parity-preserving first-layer cleanup is the mechanism of [15, Lemma 4.1]; Lemma 11 reproduces it with the full preparation, parity-preservation argument, and gate accounting needed here. The product-reflection normal form is used in [13], and the active/identity-sector decomposition of the final gate appears in [15, Section 2.3.1]. Our additional ingredient is the quantitative certificate and its incidence-based grouping. Bounded differences and its bounded-range exponential estimate are classical tools [16, 17], as is Chebyshev polynomial approximation [18]. We also use standard trace-norm duality, Schatten Hölder, spectral decomposition, and AM–GM. Their use is not asserted to be a new general technique. The needed bounded-differences statement is proved explicitly in Lemma 7.

## 1.4 Overview of the proof strategy

**From circuits to one-input sources.** For the exact theorem, parity-preserving cleanup leaves  $r \geq n/3$  variable inputs with at most one variable in each first-layer component. Writing the signed input average as a tensor product of source operators reduces the problem to bounding one full-parity coefficient. The mixedness budget controls how the trace-norm distinguishability of the two source states changes when ports are retained or discarded.

**From local retention to exponential decay.** After expanding a measurement event, independent activation coins are attached to gate events and private port events. Two sources may meet the same gate and hence depend on the same coin: multiplying their separate mean bounds would be invalid. The bounded-differences estimate (11) instead uses the source weights to control the influence of each shared coin.

A preliminary argument deletes gates whose total incident weight exceeds a threshold, but incurs an additive error depending on gate count. The exponential theorem avoids that loss by retaining the exact reflection identity

$$DBD = B - 2\Pi B - 2B\Pi + 4\Pi B\Pi, \quad D = I - 2\Pi.$$

Here  $\Pi$  is a tensor product of one-qubit rank-one projectors, and  $B$  is the corresponding port-product event. Each correction inserts rank-one operators into its incident sources. Several corrections on one source are bounded through their *joint* inserted marginal, not by multiplying separate port penalties. The insertion estimates preserve the original bounded-difference controls. A split into sparse and dense correction sets balances the expansion count against the joint norm cost, giving  $K_0 e^{-ar}$  independently of circuit size.

**From event decay to the exact gate lower bound.** Pulling a small-support observable through the third layer produces a linear combination of port-product events with controlled coefficient mass. Exactness on the final gate’s identity sector supplies a positive certificate before that gate. The argument uses no postselection and does not infer determinism merely from a nonzero projection. Applying the certificate separately to every control would introduce an uncontrolled arity factor.

Instead, a *cell* consists of controls sharing a source, a second-layer gate or idle label, and a third-layer gate or idle label. The source label includes one group for all input-free first-layer components. With  $m_2, m_3$  gates in the two intervening layers, there are at most  $(r+1)(m_2+1)(m_3+1)$  cells. Rank-one polarization within a whole cell has dimension-independent cost because the event theorem permits entanglement inside a port. The resulting inequality is

$$1 \leq [36 + 2664(r+1)(m_2+1)(m_3+1)] K_0 e^{-ar}.$$

As  $m_2, m_3 \leq M$  and  $r \geq n/3$ , it forces  $M = \exp(\Omega(n))$  without counting ancillary wires.

**Why the bounded-error argument is different.** For arbitrary first layers, expand gates containing more than a fixed number of original inputs. A correction on a  $k$ -input component has trace norm at most  $2^{-k}$  before its scalar coefficient is included. Conditioning the remaining bounded-input components produces enough one-input sources to apply the exponential theorem, and the correction cost keeps the total branch contribution exponentially small with a fixed cutoff. This proves the physical-product-event bound and its three-layer small-support consequence without using exactness.

At the final gate, a Chebyshev polynomial approximates the AND of its  $b$  control projectors, for  $b \geq 2$ , with operator-norm error at most  $2e^{-2\ell/\sqrt{b}}$  at degree  $\ell$ . The three-layer estimate bounds the associated low-support terms with a coefficient cost exponential in  $\ell$ . Taking  $\ell$  to be a sufficiently small constant multiple of  $n$  balances these terms and yields the displayed final-arity bound. Thus the exact result uses a positivity certificate and cells, whereas the bounded-error result uses polynomial approximation.

The overall method estimates selected event coefficients directly; it does not establish global Pauli-spectrum concentration or low-degree approximation of the entire circuit channel. Accordingly, the broader conclusions about other functions, learning, and state synthesis in [10, 11, 12, 13, 14, 15] are not claimed here.

## 1.5 Organization of the paper

Section 2 fixes the model and states the main theorem. Section 3 proves the source and insertion estimates, and Section 4 establishes the preliminary product-event bound and the required concentration inequality. Section 5 proves exponential decay by heavy-gate expansion. Sections 6 and 7 give cleanup and three-layer pullback, respectively. Section 8 establishes the exact final-layer certificate; Section 9 refines it using cells and completes the main proof. Section 10 treats arbitrary first layers and derives the bounded-error consequences.

## 2 Circuit model, notation, and statements

**Registers, gates, and resources.** Fix integers  $N \geq n \geq 1$ . A circuit acts on  $N$  qubits, with one copy of the classical input  $|x\rangle$ ,  $x \in \{0, 1\}^n$ , and a pure ancillary state

$$|\alpha\rangle = \bigotimes_{q=n+1}^N |\alpha_q\rangle,$$

where each  $\alpha_q$  is a unit vector. For  $N = n$  the ancillary register is absent. The circuit, ancillary state, output wire, and measurement are fixed independently of  $x$ . There are no intermediate measurements or postselection. The ancillary states may be taken to be  $|0\rangle$ , because their preparation requires only one-qubit unitaries.

A generalized controlled- $Z$  gate on a set  $S$  of at least two qubits is

$$\text{CZ} * S = I_S - 2|1^S\rangle\langle 1^S|, \quad |1^S\rangle = \bigotimes_{*q \in S} |1\rangle.$$

For any target  $t \in S$ , conjugating this gate by the Hadamard on  $t$  gives the generalized Toffoli with controls  $S \setminus \{t\}$ :

$$H_t \text{CZ} * S H_t : |z\rangle|b\rangle \mapsto |z\rangle|b \oplus \bigwedge_{*q \in S \setminus \{t\}} z_q\rangle.$$

The same conjugation gives the reverse conversion. Thus these two gate sets are equivalent under the free-local-unitary convention below.

For an integer  $d \geq 0$ , a circuit of *entangling depth at most  $d$*  has  $d$  layers of controlled- $Z$  gates, with pairwise disjoint gate supports within each layer. Arbitrary products of one-qubit unitaries may occur before, between, and after these layers; they contribute neither to entangling depth nor to size. Empty layers are allowed, so depth four means depth at most four. The *size*  $M$  is the total number of multiqubit gates. Every circuit has finitely many wires and gates, but we impose no a priori bound on  $N$  or on gate arities as functions of  $n$ . There is one designated output qubit  $o$ , which may be an original input wire. Every other wire may contain arbitrary final garbage; input preservation and ancillary uncomputation are not required.

**Reflection normal form.** The conversion to product reflections preserves the number and supports of the multiqubit gates, but retains a final local unitary. For  $d = 0$  the circuit is already a product of one-qubit unitaries. For  $d \geq 1$ , write

$$C = L_d D_d L_{d-1} D_{d-1} \cdots L_1 D_1 L_0, \quad K_j = L_j L_{j-1} \cdots L_0,$$

where each  $L_j$  is a product of one-qubit unitaries and each  $D_j$  is a disjoint controlled- $Z$  layer. Then

$$R_j = K_{j-1}^* D_j K_{j-1}, \quad C = K_d R_d R_{d-1} \cdots R_1.$$

This identity follows by cancellation of consecutive  $K_j$  factors, or by repeated use of  $DL = L(L^*DL)$ . Because  $K_{j-1}$  is local, each gate in  $R_j$  has the same support  $S$  as its original gate and has the form

$$D_S = I_S - 2\Pi_S, \quad \Pi_S = \bigotimes_{q \in S} |\theta_q\rangle\langle \theta_q|, \quad (1)$$

for one-qubit unit vectors  $\theta_q$ . Conversely, choose one-qubit unitaries  $V_q$  with  $V_q|1\rangle = |\theta_q\rangle$ . For  $V_S = \bigotimes_{q \in S} V_q$ ,

$$D_S = V_S \text{CZ}_S V_S^*.$$

Disjointness allows these local conjugations to be performed simultaneously within each layer, and neighboring local layers can be merged. This proves equivalence in both directions without changing entangling depth or multiqubit-gate count. The remaining  $K_d$  may be omitted for output statistics if the one-qubit measurement observable  $A_o$  is replaced by  $K_{d,o}^* A_o K_{d,o}$ , with  $K_{d,o}$  its output-qubit factor.

**Parity coefficients, events, and norms.** Put  $[n] = \{1, \dots, n\}$  and  $\chi(x) = (-1)^{x_1 + \cdots + x_n}$ . All input averages are uniform. For  $q : \{0, 1\}^n \rightarrow \mathbb{C}$ , its full-parity coefficient is

$$\hat{q}([n]) = \mathbb{E}_x \chi(x) q(x) = 2^{-n} \sum_{x \in \{0, 1\}^n} \chi(x) q(x).$$

For a state  $\rho_x$  at a specified circuit stage, write  $\langle B \rangle_x = \text{Tr}(\rho_x B)$ . In such expectations, an operator specified on only some wires is tensored with identity on the remaining wires.

A *physical-qubit product event*, or simply a *product event* when no other partition is specified, is  $E = \bigotimes_q E_q$ , where each  $E_q$  is an identity or a one-qubit rank-one orthogonal projector. Later statements explicitly allow the larger class with  $0 \leq E_q \leq I_q$ . For a specified partition into blocks, a *port-product effect* is instead a tensor product  $\bigotimes_j E_j$  with  $0 \leq E_j \leq I_j$  on each block. Such a factor need not be a product over its physical qubits. Sections 4 and 5 specify the source–second-layer ports for their event estimates; these estimates must not be confused with the physical-qubit product-event statement in Section 10.

For a linear map  $T : \mathcal{H} \rightarrow \mathcal{K}$  between specified finite-dimensional Hilbert spaces, we use the unnormalized norms

$$\|T\|_1 = \text{Tr} \sqrt{T^* T}, \quad \|T\|_\infty = \sup_{\|v\|=1} \|Tv\|, \quad \|T\|_2 = \sqrt{\text{Tr}(T^* T)}.$$

A *contraction* has operator norm at most one; a *positive contraction*, or measurement effect, satisfies  $0 \leq T \leq I$ . The identity-extension convention for expectations does not extend the space on which a standalone trace norm or Hilbert–Schmidt norm is evaluated.

**Output conventions and the main theorem.** Let  $|\Psi_x\rangle = C(|x\rangle \otimes |\alpha\rangle)$  and  $\rho_x = |\Psi_x\rangle\langle\Psi_x|$  be the full output state. A binary measurement on  $o$  consists of effects  $E_+, E_- \geq 0$  with  $E_+ + E_- = I_o$ . Label its outcomes  $+1$  for even parity and  $-1$  for odd parity, and put

$$A_o = E_+ - E_- = 2E_+ - I_o.$$

Thus  $A_o$  is a one-qubit Hermitian contraction and  $\text{Tr}(\rho_x A_o)$  is the mean of the signed measurement outcome. Conversely, every one-qubit Hermitian contraction defines such a measurement by  $E_\pm = (I_o \pm A_o)/2$ . For a general binary measurement the outcomes are these labels, not necessarily the eigenvalues of  $A_o$ .

We say that  $C$  *computes parity exactly* if

$$\text{Tr}(\rho_x A_o) = \chi(x) \quad \text{for every } x \in \{0, 1\}^n. \quad (2)$$

Because  $n \geq 1$ , both parity values occur. Attaining each of the extreme expectations  $+1$  and  $-1$  forces both values to occur in the spectrum of  $A_o$ . Since  $o$  is a qubit,  $A_o$  is therefore a Hermitian unitary with one eigenvalue of each sign. In particular, allowing a general binary measurement does not enlarge this exact computation model.

In the exact case, let  $|a_\pm\rangle$  be fixed unit eigenvectors of  $A_o$ . For  $y = \chi(x)$ , the projector  $P_y = \frac{I_o + y A_o}{2} \otimes I_{\text{rest}}$  satisfies

$$\|(I - P_y)|\Psi_x\rangle\|^2 = \frac{1 - y \text{Tr}(\rho_x A_o)}{2}.$$

Exactness is consequently equivalent to

$$|\Psi_x\rangle = |a_{\chi(x)}\rangle * o \otimes |\gamma_x\rangle * \text{rest}$$

for a unit vector  $\gamma_x$  that may depend arbitrarily on  $x$ . The purity used here follows from pure initialization and unitary evolution. No preservation of the other registers is asserted.

For a general binary measurement, define the uniform-input parity correlation by

$$c(C, A_o) = \mathbb{E}_x \chi(x) \text{Tr}(\rho_x A_o). \quad (3)$$

On input  $x$ , the probability of the correct outcome is  $[1 + \chi(x) \text{Tr}(\rho_x A_o)]/2$ . Hence the average success probability is exactly

$$p_{\text{succ}}(C, A_o) = \frac{1 + c(C, A_o)}{2}.$$

In particular, uniform-input success at least  $1/2 + \varepsilon$  implies  $c(C, A_o) \geq 2\varepsilon$ . A worst-case success guarantee implies the same uniform-input guarantee, but the converse is not assumed.

**Theorem 1** (Exact depth-four lower bound). *There exist an absolute constant  $c_* > 0$  and an absolute integer  $n_0 \geq 1$  such that, for every  $n \geq n_0$  and every circuit in the model above of entangling depth at most four that computes  $n$ -bit parity exactly, its multiqubit-gate count satisfies*

$$M \geq e^{c_* n}.$$

*The constants are independent of the finite number of product-initialized ancillary wires and of all gate arities. In particular, exact parity has no polynomial-size depth-four QAC circuit family, even when only multiqubit gates are counted.*

The proof is completed in Section 9, using the exponential event estimate and parity-preserving cleanup from Sections 5 and 6. It permits arbitrary garbage and also covers an output qubit that is an original input wire.

### 3 A local source lemma

After one disjoint gate layer, the state is a tensor product over gate components, including untouched singleton wires. A *source* in this section is one such component with exactly one variable input and any finite number of fixed pure product ancillas. Denote its variable-input wire by  $d$ . Partition the source wires into nonempty *ports*, indexed by  $\mathcal{J} = \{0, 1, \dots, m\}$ . The *owner* port 0 contains  $d$ ; all other ports contain only ancillary wires. This partition is arbitrary and will later be chosen from second-layer incidences. For  $S \subseteq \mathcal{J}$ , the notation  $\text{Tr}_{S^c}$  means tracing all wires in ports indexed by  $\mathcal{J} \setminus S$ . In particular, if  $S$  contains only ancillary-port indices, its complement includes the owner.

Using the reflection normal form and a local change of basis on  $d$ , write the source gate as

$$G = I - 2|1\rangle\langle 1|_d \otimes \Pi_{\text{anc}}, \quad \Pi_{\text{anc}} = \bigotimes_{a \in \mathcal{A}} |v_a\rangle\langle v_a|,$$

where  $\mathcal{A}$  is the set of ancillary wires in the source. All vectors  $v_a$  and  $\alpha_a$  below are unit vectors. The two possible data inputs  $u_0, u_1$  form an orthonormal basis, and the ancillary input is  $\alpha = \bigotimes_{a \in \mathcal{A}} \alpha_a$ . Local unitaries after  $G$  conjugate the port marginals and preserve their eigenvalues and the retained trace norms. In the insertion lemma below they can also be absorbed into the inserted operator without changing its rank or norm, so we omit them here.

An untouched singleton is covered by taking no ancillas and using the formal one-qubit reflection  $G = Z_d$ . If its actual output vectors are  $w_0, w_1$ , set  $u_x = Z_d w_x$ , so  $G u_x = w_x$ . This is an analytic representation, not an added multiqubit gate. Empty tensor products are scalars on a one-dimensional space.

Put

$$\psi_x = G(u_x \otimes \alpha), \quad F = \frac{|\psi_0\rangle\langle\psi_0| - |\psi_1\rangle\langle\psi_1|}{2}, \quad \bar{\rho} = \frac{|\psi_0\rangle\langle\psi_0| + |\psi_1\rangle\langle\psi_1|}{2}.$$

The vectors  $\psi_0, \psi_1$  are orthonormal because  $G$  is unitary. Thus  $F = F^*$ ,  $\text{Tr } F = 0$ , and  $\|F\|_1 = 1$ . For each port  $j$ , define

$$\bar{\rho}_j = \text{Tr}_{\mathcal{J} \setminus \{j\}} \bar{\rho}, \quad \lambda_j = \lambda_{\max}(\bar{\rho}_j), \quad h_j = 1 - \lambda_j.$$

Set  $C_s = 1/2 + 2/e$ .

**Lemma 2** (Mixedness and random retention). *For a source as above:*

- (i)  $h_0 = 1/2$  and  $\sum_{j \in \mathcal{J}} h_j \leq C_s$ .
- (ii) There is a function  $f : 2^{\mathcal{J}} \rightarrow [0, 1]$  such that

$$\|\text{Tr}_{S^c} F\|_1^2 \leq f(S) \quad (S \subseteq \mathcal{J}).$$

If each port is retained independently with probability  $1/2$ , then  $\mathbb{E}f \leq 3/4$ .

- (iii) For every  $S \subseteq \mathcal{J}$  and every  $j \in \mathcal{J}$ ,

$$|f(S \triangle \{j\}) - f(S)| \leq 4h_j,$$

where  $\triangle$  denotes symmetric difference.

These assertions allow arbitrary finite port widths and arbitrarily many fixed pure product ancillas in the owner port.

*Proof.* Define

$$\delta = |\langle 1|u_0\rangle|^2 - |\langle 1|u_1\rangle|^2, \quad |\delta| \leq 1.$$

For each port  $j$ , let  $\mathcal{A}_j$  be its ancillary wires and set

$$\alpha^{(j)} = \bigotimes_{a \in \mathcal{A}_j} \alpha_a, \quad \Pi^{(j)} = \bigotimes_{a \in \mathcal{A}_j} |v_a\rangle\langle v_a|.$$

The data wire is excluded from both products when  $j = 0$ . For  $j \neq 0$  put

$$p_j = \langle \alpha^{(j)} | \Pi^{(j)} | \alpha^{(j)} \rangle = \prod_{a \in \mathcal{A}_j} |\langle v_a | \alpha_a \rangle|^2, \quad t_0 = \langle \alpha^{(0)} | \Pi^{(0)} | \alpha^{(0)} \rangle, \quad P = t_0 \prod_{j \neq 0} p_j.$$

All these overlap probabilities lie in  $[0, 1]$ ; empty products are one.

Since the uniform mixture of  $u_0, u_1$  is  $I_d/2$ , averaging gives

$$\bar{\rho} = \frac{1}{2} |0\rangle\langle 0| \otimes |\alpha\rangle\langle \alpha| + \frac{1}{2} |1\rangle\langle 1| \otimes R_{\text{anc}} |\alpha\rangle\langle \alpha| R_{\text{anc}}, \quad R_{\text{anc}} = I - 2\Pi_{\text{anc}}. \quad (4)$$

On the owner port, the data-zero block is a pure ancillary projector of weight  $1/2$ . The orthogonal data-one block is  $1/2$  times a density operator. Every eigenvalue is therefore at most  $1/2$ , and the data-zero block attains  $1/2$ . Hence  $\lambda_0 = 1/2$ , even when the owner contains ancillas.

If  $P = 0$ , then  $\Pi_{\text{anc}} \alpha = 0$ , so  $G$  acts as the identity on the two-dimensional input subspace spanned by  $u_x \otimes \alpha$ . All ancillary-only ports remain pure. Taking  $f(S) = \mathbf{1}_{\{0 \in S\}}$  gives the exact squared retained norm: its mean is  $1/2$ , ancillary-port changes are zero, and the owner change is  $1 = 2h_0 \leq 4h_0$ . This proves all assertions in this case. Henceforth assume  $P > 0$ , so all  $p_j$  and  $t_0$  are positive.

For an ancillary-only port  $j$ , put  $r_j = P/p_j \in (0, 1]$  and  $R_j = I_j - 2\Pi^{(j)}$ . Tracing (4) gives

$$\bar{\rho}_j = (1 - r_j/2) |\alpha^{(j)}\rangle\langle \alpha^{(j)}| + (r_j/2) R_j |\alpha^{(j)}\rangle\langle \alpha^{(j)}| R_j.$$

Indeed, the reduced reflected pure state is  $(1 - r_j) |\alpha^{(j)}\rangle\langle \alpha^{(j)}| + r_j R_j |\alpha^{(j)}\rangle\langle \alpha^{(j)}| R_j$ , as follows by expanding the reflection and tracing the other ancillas. The marginal has rank at most two, so its two possibly nonzero eigenvalues are  $1 - h_j, h_j$ , with  $0 \leq h_j \leq 1/2$ . For a mixture  $(1 - t)|u\rangle\langle u| + t|v\rangle\langle v|$  of unit vectors, the product of these two eigenvalues is  $t(1 - t)(1 - |\langle u|v\rangle|^2)$ : restrict to their span

and use the orthonormal basis consisting of  $u$  and the normalized component of  $v$  orthogonal to  $u$ ; if the span is one-dimensional, both sides are zero. Here  $\langle \alpha^{(j)} | R_j | \alpha^{(j)} \rangle = 1 - 2p_j$ , giving

$$h_j(1 - h_j) = \frac{r_j}{2} \left(1 - \frac{r_j}{2}\right) [1 - (1 - 2p_j)^2] = P(1 - p_j)(2 - r_j). \quad (5)$$

Since  $h_j(1 - h_j) \leq h_j$  and  $2 - r_j \geq 1$ , this implies  $h_j \geq P(1 - p_j)$ . For the other direction, the variational characterization of the largest eigenvalue gives

$$1 - h_j = \lambda_j \geq \langle \alpha^{(j)} | \bar{\rho}_j | \alpha^{(j)} \rangle = 1 - 2P(1 - p_j).$$

Thus

$$P(1 - p_j) \leq h_j \leq 2P(1 - p_j). \quad (6)$$

Writing  $Q = \prod_{j \neq 0} p_j$ , the inequality  $1 - p_j \leq -\log p_j$  and  $P = t_0 Q \leq Q$  yield

$$\sum_{j \neq 0} h_j \leq 2P \sum_{j \neq 0} (1 - p_j) \leq -2P \log Q \leq -2P \log P \leq \frac{2}{e}.$$

The last step maximizes  $-2P \log P$  on  $(0, 1]$ . With  $h_0 = 1/2$ , this proves (i), including the empty ancillary-port collection.

We next compute the signed marginal when the owner is erased. For retained ancillary-only ports  $A \subseteq \mathcal{J} \setminus \{0\}$ , put

$$P_A = \prod_{j \in A} p_j, \quad P_I = t_0 \prod_{\substack{j \notin A \\ j \neq 0}} p_j, \quad P = P_A P_I.$$

Here  $P_I$  includes the overlap probabilities of the owner ancillas. Let  $\alpha_A = \bigotimes_{j \in A} \alpha^{(j)}$ ,  $\rho_A = |\alpha_A\rangle\langle\alpha_A|$ ,  $\Pi_A = \bigotimes_{j \in A} \Pi^{(j)}$ , and  $R_A = I_A - 2\Pi_A$ . The data-zero and data-one diagonal coefficients of the signed input are  $-\delta/2$  and  $\delta/2$ , respectively. Hence

$$\text{Tr}_d F = \frac{\delta}{2} (R_{\text{anc}} |\alpha\rangle\langle\alpha| R_{\text{anc}} - |\alpha\rangle\langle\alpha|).$$

Expanding the reflection and tracing all erased ancillas gives

$$\begin{aligned} \text{Tr}_{\text{erased ancillas}} (R_{\text{anc}} |\alpha\rangle\langle\alpha| R_{\text{anc}}) &= \rho_A - 2P_I \Pi_A \rho_A - 2P_I \rho_A \Pi_A + 4P_I \Pi_A \rho_A \Pi_A \\ &= \rho_A + P_I (R_A \rho_A R_A - \rho_A). \end{aligned}$$

Consequently,

$$\text{Tr}_{A^c} F = \frac{\delta P_I}{2} (R_A \rho_A R_A - \rho_A).$$

For unit vectors  $u, v$ , the trace norm of the difference of their pure-state projectors is

$$\| |u\rangle\langle u| - |v\rangle\langle v| \|_* = 2\sqrt{1 - |\langle u|v\rangle|^2}.$$

Indeed, on their span the difference has trace zero and eigenvalues  $\pm\sqrt{1 - |\langle u|v\rangle|^2}$ ; the coincident-state case gives zero. Using  $\langle \alpha_A | R_A | \alpha_A \rangle = 1 - 2P_A$ , we obtain

$$\begin{aligned} D(A) &:= \|\text{Tr} * A^c F\|_1 = 2|\delta| P_I \sqrt{P_A(1 - P_A)}, \\ D(A)^2 &= 4\delta^2 P^2 (P_A^{-1} - 1). \end{aligned} \quad (7)$$

Division by  $P_A$  is legitimate because  $P > 0$ . If  $A = \emptyset$ , its complement is every source port,  $P_A = 1$ , and  $D(A) = |\text{Tr } F| = 0$ , consistently with this formula.

Define

$$f(S) = \begin{cases} 1, & 0 \in S, \\ D(S)^2, & 0 \notin S. \end{cases}$$

Every reduced signed state is half the difference of two density operators, so its trace norm is at most one by the triangle inequality. Thus  $0 \leq \|\text{Tr}_{S^c} F\|_1^2 \leq f(S) \leq 1$ . In the owner-erased case  $f$  is the exact squared norm; in the owner-retained case it is only an upper bound.

Conditioned on owner erasure, independent half-retention of the other ports gives

$$\mathbb{E}_A D(A)^2 = 4\delta^2 t_0^2 Q \left[ \prod_{j \neq 0} \frac{1+p_j}{2} - Q \right].$$

For example, this follows directly from  $\mathbb{E}_A P_A^{-1} = \prod_{j \neq 0} \frac{1+p_j^{-1}}{2}$ . For numbers in  $[0, 1]$ ,

$$\prod_j \frac{1+p_j}{2} \leq \frac{1 + \prod_j p_j}{2}.$$

The empty-product case is equality, and the induction step uses

$$\frac{1+ab}{2} - \frac{(1+a)(1+b)}{4} = \frac{(1-a)(1-b)}{4} \geq 0.$$

It follows that  $\mathbb{E}_A D(A)^2 \leq 2\delta^2 t_0^2 Q(1-Q) \leq 1/2$ . The owner's independent retention coin therefore gives

$$\mathbb{E} f = \frac{1}{2} + \frac{1}{2} \mathbb{E}_A D(A)^2 \leq \frac{3}{4},$$

proving (ii).

Finally, suppose the owner is erased and  $j \neq 0$  is not yet in  $A$ . By (7), adding  $j$  changes the majorant by

$$D(A \cup \{j\})^2 - D(A)^2 = 4\delta^2 \frac{P^2}{P_A p_j} (1-p_j) \leq 4P(1-p_j) \leq 4h_j.$$

The change is nonnegative; the first inequality uses  $P/(P_A p_j) \leq 1$ , the product of overlaps of the still-erased ancillas, and the last uses (6). The same absolute bound holds when  $j$  is erased instead. If the owner is already retained, an ancillary-port change leaves  $f = 1$  unchanged. Adding or erasing the owner changes  $f$  by at most  $1 = 2h_0 \leq 4h_0$ . This proves (iii).  $\square$

*Remark 3.* The constant  $3/4$  is sharp. Take  $u_x = |x\rangle$  and one ancilla in  $|0\rangle$ , apply CNOT from the input to the ancilla, and place the two wires in separate ports. This is directly a permitted source reflection:

$$\text{CNOT} = I - 2|1\rangle\langle 1|_d \otimes |-\rangle\langle -|, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

It has  $\delta = -1$ ,  $t_0 = 1$ , and  $p_1 = 1/2$ , and its signed output is

$$F = \frac{|00\rangle\langle 00| - |11\rangle\langle 11|}{2}.$$

For  $S = \emptyset, \{0\}, \{1\}, \{0, 1\}$ , respectively, the squared retained trace norms are  $0, 1, 1, 1$ . The majorant above has these same values, so its mean is exactly  $3/4$ . Indeed, any majorant of the true squared retained norm must have mean at least  $3/4$  in this example.

The example also shows why separate insertion penalties cannot be multiplied. Here  $\lambda_0 = \lambda_1 = 1/2$ , but

$$|\text{Tr}[(|0\rangle\langle 0| \otimes |0\rangle\langle 0|)F]| = \frac{1}{2} > \lambda_0\lambda_1.$$

The joint marginal, rather than a product of single-port bounds, is needed for simultaneous insertions.

**Lemma 4** (Rank-one insertion). *For a one-input source and any operator  $B$  of rank at most one on port  $j$ , put  $K = \text{Tr}_j[(B \otimes I)F]$ . Then*

$$\|K\|_1 \leq \|B\|_\infty \lambda_j.$$

*In particular, a rank-one contraction gives  $\|K\|_1 \leq \lambda_j$ . For a union  $J \subseteq \mathcal{J}$  of ports, take  $B$  of rank at most one on that union and put  $K = \text{Tr}_J[(B \otimes I)F]$ . The same statement holds with the joint marginal  $\bar{\rho}_J = \text{Tr}_{J^c} \bar{\rho}$  and  $\lambda_J = \lambda_{\max}(\bar{\rho}_J)$ .*

*If  $L, R, T$  are contractions on the remaining wires, then  $\|LKR\|_1 \leq \|B\|_\infty \lambda_j$  and  $|\text{Tr}(TK)| \leq \|B\|_\infty \lambda_j$ . Further partial traces cannot increase the trace-norm bound. For insertion on a union  $J$ , these consequences hold with  $\lambda_J$  in place of  $\lambda_j$ . All these assertions concern unnormalized operators.*

*Proof.* Define the isometry  $W : \mathbb{C}^2 \rightarrow \mathcal{H}_{\text{source}}$  by  $W|x\rangle = \psi_x$ . Let  $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$  on its logical input. Then

$$F = WZW^*/2, \quad \bar{\rho} = WW^*/2.$$

For a unit vector  $u$  on the inserted port define  $K_u = (\langle u| \otimes I)W$ . Cyclicity of trace gives

$$\|K_u\|_2^2 = \text{Tr}[(|u\rangle\langle u| \otimes I)WW^*] = 2\langle u|\bar{\rho}_j|u\rangle \leq 2\lambda_j.$$

For  $B = |u\rangle\langle v|$  with  $u, v$  unit, direct partial tracing gives

$$K = K_v Z K_u^*/2.$$

Schatten Hölder and  $\|K_v Z\|_2 = \|K_v\|_2$  yield

$$\|K\|_1 \leq \frac{1}{2} \|K_v\|_2 \|K_u\|_2 \leq \lambda_j.$$

Every nonzero rank-one  $B$  has the form  $s|u\rangle\langle v|$  with  $s = \|B\|_\infty$  and  $u, v$  unit; scaling proves the stated bound. For  $B = 0$  it is immediate. Nothing requires  $B$  or  $K$  to be Hermitian.

The same calculation treats a union of ports as one tensor factor. The remaining-wire assertions follow from  $\|LKR\|_1 \leq \|L\|_\infty \|K\|_1 \|R\|_\infty$  and trace-norm duality. For completeness, partial-trace contractivity also holds without Hermiticity:

$$\|\text{Tr}_{\mathcal{B}} X\|_1 = \sup_{\|Y\|_\infty \leq 1} |\text{Tr}[(I_{\mathcal{B}} \otimes Y)X]| \leq \|X\|_1.$$

The insertion argument uses only that  $W$  is an isometry. The special mixedness values and budget proved above still require the one-reflection, product-initialized source structure.  $\square$

## 4 Two-layer product events on arbitrary incidence graphs

**Theorem 5** (Product-event concentration). *Consider a circuit with two multiqubit layers, each consisting of pairwise wire-disjoint generalized controlled-Z gates, together with arbitrary one-qubit gates and fixed pure product ancillas. Suppose the first layer has  $r \geq 1$  input-containing components,*

including untouched input singletons, each containing exactly one variable input. A port is a nonempty intersection of a first-layer source with a second-layer gate, or the nonempty set of that source's remaining idle wires. Let  $E$  be a tensor product of positive contractions over these ports; an individual port effect need not factor across its physical qubits.

Let  $\rho_x$ , for  $x \in \{0,1\}^r$ , be the state immediately before the second multiqubit layer, including the preceding one-qubit gates, and absorb any final one-qubit gates into  $E$ . Let  $U_2$  be the product of the  $m \geq 0$  second-layer multiqubit gates, and set

$$q(x) = \text{Tr}(EU_2\rho_x U_2^*), \quad \hat{q}([r]) = 2^{-r} \sum_{x \in \{0,1\}^r} (-1)^{x_1 + \dots + x_r} q(x).$$

Then, for every real  $w \geq 1$ ,

$$|\hat{q}([r])| \leq \Delta_r(m, w) := 8me^{-w} + \left(\frac{7}{8}\right)^{r/2} + \exp\left(-\frac{r}{512C_s w}\right). \quad (8)$$

There is no degree, treewidth, or ancilla-width restriction on the incidence graph between the two layers. All input-free first components may be grouped into one resource source. More generally, the same bound holds if that source is replaced by an arbitrary input-independent density operator  $\sigma$ , tensor-independent of the  $r$  input sources. Ports on the resource are defined after this grouping.

*Proof.* The one-qubit gates absorbed at the two boundaries act within ports; they preserve the source estimates, positivity, contractivity, and the event's port factorization. For each input source  $i$ , write  $\rho_{i,b}$  for its state on input  $b \in \{0,1\}$  and  $F_i = (\rho_{i,0} - \rho_{i,1})/2$ . Up to reordering tensor factors,

$$\rho_x = \sigma \otimes \bigotimes_{i=1}^r \rho_{i,x_i}, \quad \Sigma := 2^{-r} \sum_x (-1)^{x_1 + \dots + x_r} \rho_x = \sigma \otimes \bigotimes_{i=1}^r F_i.$$

Each  $F_i$  satisfies Section 3, including any final port-local rotations, and  $\|F_i\|_1 = 1$ . If no resource wires are present,  $\sigma$  is the scalar 1 on a one-dimensional space. Thus  $\hat{q}([r]) = \text{Tr}[(U_2^* E U_2) \Sigma]$ . The resource may be internally entangled across all its ports.

Represent each second-layer gate by a real vertex. For every source, including the resource, with nonempty idle wires, add one private virtual vertex carrying the effect on its idle port. Include identity effects on unmeasured wires. Every wire belongs to exactly one vertex, and a vertex meets each source in at most one port. For a real vertex  $g$ , write

$$P_g = P_{\text{res},g} \otimes \bigotimes_{i: i \sim g} P_{ig}, \quad D_g = I - 2\Pi_g, \quad \Pi_g = \Pi_{\text{res},g} \otimes \bigotimes_{i: i \sim g} \Pi_{ig}.$$

Absent factors are omitted. Here  $i \sim g$  means that source  $i$  meets  $g$ ; every present  $\Pi_{ig}$ , and likewise  $\Pi_{\text{res},g}$ , is a rank-one orthogonal projector. The pulled-back effect at  $g$  is  $D_g P_g D_g$ . All real and virtual vertex effects act on disjoint wire sets, so their tensor product is  $U_2^* E U_2$ .

**Delete heavy gates.** For every input source  $i$  and vertex  $g$ , let  $(\lambda_{ig}, h_{ig})$  be its original port quantities from Section 3. Set  $h_{ig} = 0$  and  $\lambda_{ig} = 1$  when the port is absent. In particular,

$$\sum_g h_{ig} \leq C_s, \quad 0 \leq h_{ig} \leq \frac{1}{2}.$$

The individual bound uses  $h_0 = 1/2$  for the owner and the rank-at-most-two ancillary marginals proved in Section 3. Resource ports do not contribute to these weights. Call a real vertex heavy if  $\sum_i h_{ig} > w$ . Deleting its gate replaces  $D_g P_g D_g$  by  $P_g$ , with difference

$$D_g P_g D_g - P_g = -2\Pi_g P_g - 2P_g \Pi_g + 4\Pi_g P_g \Pi_g. \quad (9)$$

After removing the scalar coefficients  $-2, -2, 4$ , each of the three operators on the right factors over source ports. Its factor on input port  $(i, g)$  is, respectively,

$$B_{ig}^{(1)} = \Pi_{ig} P_{ig}, \quad B_{ig}^{(2)} = P_{ig} \Pi_{ig}, \quad B_{ig}^{(3)} = \Pi_{ig} P_{ig} \Pi_{ig}.$$

Each has rank at most one and norm at most one, even if  $P_{ig}$  has high rank; zero factors are allowed. The resource factors have the same forms and are also contractions.

To bound one deletion in a telescoping sequence, let  $M_{-g}$  be the tensor product of the effects on all other vertices. An earlier deletion has merely replaced some of these effects by their  $P$  factors. They remain positive contractions, so  $\|M_{-g}\|_\infty \leq 1$ . For a coefficient-free term  $\ell$ , trace the wires at  $g$  first:

$$K_i^{(\ell)} = \text{Tr}_{(i,g)}[(B_{ig}^{(\ell)} \otimes I)F_i], \quad \|K_i^{(\ell)}\|_1 \leq \lambda_{ig}$$

by Lemma 4; for an absent port take  $K_i^{(\ell)} = F_i$ . The resource factor similarly satisfies

$$\|K_{\text{res}}^{(\ell)}\|_1 = \|\text{Tr}_{\text{res},g}[(B_{\text{res},g}^{(\ell)} \otimes I)\sigma]\|_1 \leq 1.$$

Indeed, for any inserted contraction  $T$  and any remaining-wire contraction  $R$ , trace-norm duality gives  $|\text{Tr}[(T \otimes R)\sigma]| \leq \|\sigma\|_1 = 1$ . This argument allows non-Hermitian  $T$  and arbitrary internal resource entanglement; an absent resource port leaves  $\sigma$  unchanged.

The contribution of term  $\ell$  is the pairing of  $M_{-g}$  with  $K_{\text{res}}^{(\ell)} \otimes \bigotimes_i K_i^{(\ell)}$ . Trace-norm multiplicativity and duality therefore bound its magnitude by  $\prod_i \lambda_{ig}$ . Restoring the coefficients in (9), the change in the full-parity coefficient is at most

$$(2 + 2 + 4) \prod_i \lambda_{ig} \leq 8 \exp\left(-\sum_i h_{ig}\right) \leq 8e^{-w},$$

where  $\lambda_{ig} = 1 - h_{ig} \leq e^{-h_{ig}}$ . Every deletion uses the original  $F_i$  and its original port weights: earlier deletions change only  $M_{-g}$ . Thus no penalties for different ports of the same source are being multiplied. Telescoping over at most  $m$  heavy gates costs at most  $8me^{-w}$ .

After deleting a heavy gate, split its product effect  $P_g$  into separate private virtual vertices, one per nonempty source port, including its resource port when present. This is a refactorization of the effect, not a change to any source or source port. Every surviving real vertex has  $\sum_i h_{ig} \leq w$ . A private vertex meets at most one input source and has column weight at most  $1/2 \leq w$ ; a resource-only vertex has column weight zero. Row weights remain unchanged.

**Expand random activations.** Let  $\mathcal{V}$  be the resulting finite vertex set. Its effect  $E_g$  is  $D_g P_g D_g$  at a surviving real vertex and is the corresponding single-port factor at a private vertex. Since  $0 \leq E_g \leq I_g$ , write

$$E_g = \frac{I_g + A_g}{2}, \quad A_g = 2E_g - I_g, \quad A_g = A_g^*, \quad \|A_g\|_\infty \leq 1.$$

Choose a random subset  $A \subseteq \mathcal{V}$  by independently including every vertex with probability  $1/2$ . Disjointness gives the exact operator identity

$$\bigotimes_{g \in \mathcal{V}} E_g = \mathbb{E}_A \left[ \left( \bigotimes_{g \in A} A_g \right) \otimes \left( \bigotimes_{g \notin A} I_g \right) \right].$$

A deleted gate is thus represented by separate independent coins for its private factors, rather than a shared coin. This is an analytic expansion of the effect, not a random operation in the circuit.

Let  $\mathcal{V}_i$  be the set of vertices meeting source  $i$ , identifying its ports with these vertices. In the term for  $A$ , trace all inactive wires. The remaining signed operator is

$$\sigma_A \otimes \bigotimes_{i=1}^r \text{Tr}_{\mathcal{V}_i \setminus A} F_i,$$

where  $\sigma_A$  is the resource marginal on its active ports and  $\|\sigma_A\|_1 = 1$ , also when those ports are empty. The active vertex operators may couple different sources but their tensor product still has operator norm at most one. The triangle inequality, trace-norm multiplicativity, and duality give

$$|\widehat{q}_{\text{deleted}}([r])| \leq \mathbb{E}_A \prod_{i=1}^r D_i(A), \quad D_i(A) = \|\text{Tr}_{\mathcal{V}_i \setminus A} F_i\|_1. \quad (10)$$

Here  $q_{\text{deleted}}$  is the event probability after the heavy second-layer gates have been removed. Resource entanglement causes no additional factor, because every resource marginal is a density operator. In particular, this argument does not factor that marginal over resource ports.

**Handle the dependencies.** Regard the source majorant  $f_i$  in Lemma 2 as a function on subsets of  $\mathcal{V}_i$ , and put

$$H(A) = \sum_{i=1}^r f_i(A \cap \mathcal{V}_i).$$

Each source has at most one port per vertex, so its port-retention coins are independent and fair. Different sources can share a vertex coin. For example, take exactly two input sources, both untouched singletons, whose wires meet one second-layer gate. Its column weight is 1, so it survives for every  $w \geq 1$ , and both source majorants equal that gate's activation indicator. In that example  $\mathbb{E}(f_1 f_2) = 1/2$ , whereas  $(\mathbb{E} f_1)(\mathbb{E} f_2) = 1/4$ . No independence between the source majorants is assumed. Applying Lemma 2 separately and using linearity gives  $\mathbb{E} H \leq 3r/4$ .

Changing vertex  $g$  changes  $H$  by at most  $c_g = 4 \sum_i h_{ig}$ . The row and column bounds imply

$$\max_g c_g \leq 4w, \quad \sum_g c_g \leq 4C_s r, \quad \sum_g c_g^2 \leq 16C_s r w.$$

The last step uses  $\sum_g c_g^2 \leq (\max_g c_g) \sum_g c_g$ . There is a positive coordinate bound because  $r \geq 1$  and every source has an owner of weight  $1/2$ . Since  $\mathbb{E} H \leq 3r/4$ , the event  $H \geq 7r/8$  is an upward deviation of at least  $r/8$ . Lemma 7 therefore gives

$$\Pr[H \geq 7r/8] \leq \exp\left(-\frac{2(r/8)^2}{16C_s r w}\right) = \exp\left(-\frac{r}{512C_s w}\right). \quad (11)$$

On the complementary event, AM–GM and Lemma 2 give

$$\prod_i D_i(A) \leq \sqrt{\prod_i f_i(A \cap \mathcal{V}_i)} \leq (H(A)/r)^{r/2} \leq (7/8)^{r/2}.$$

On the exceptional event, every  $D_i(A) \leq 1$ , so the product is at most one. Splitting the expectation in (10) between these two events and adding the deletion cost proves (8).  $\square$

*Remark 6* (The heavy gates can be counted by weight). Let  $N_{\text{heavy}}$  be the number of deleted gates. Disjointness of the original ports and the source row budgets give

$$wN_{\text{heavy}} \leq \sum_{\text{heavy } g} \sum_i h_{ig} \leq C_s r, \quad N_{\text{heavy}} \leq m.$$

(The first inequality is strict if  $N_{\text{heavy}} > 0$ .) Thus the first term in (8) may be replaced by  $8 \min\{m, C_s r/w\} e^{-w}$ . In particular, using  $C_s r/w$  and choosing  $w = \sqrt{r}$  gives the gate-count-independent bound

$$|\hat{q}([r])| \leq 8C_s \sqrt{r} e^{-\sqrt{r}} + (7/8)^{r/2} + e^{-\sqrt{r}/(512C_s)}.$$

This has stretched-exponential decay in  $r$ , not the  $e^{-ar}$  decay established in Section 5.

Even the weaker bound (8), with  $w = \sqrt{r}$ , suffices to exclude polynomial-size exact depth-four families when combined with the cleanup and cell certificate in Sections 6 and 9: cleanup leaves  $r \geq n/3$ , the cell-certificate prefactor is polynomial in  $n$  for a polynomial-size family, and every term of that prefactor times  $\Delta_r(m, \sqrt{r})$  tends to zero. The full exponential gate-count lower bound uses the next section's stronger event estimate.

**Lemma 7** (Bounded differences, stated explicitly). *Let  $X_1, \dots, X_M$  be independent random variables, each taking values in a finite nonempty set  $\mathcal{X}_g$ , and let  $H : \prod_{g=1}^M \mathcal{X}_g \rightarrow \mathbb{R}$ . Suppose that  $c_g \geq 0$  and  $|H(x) - H(x')| \leq c_g$  whenever  $x, x'$  differ only in coordinate  $g$ . Writing  $H$  also for  $H(X_1, \dots, X_M)$ , for  $t \geq 0$  and  $\sum_g c_g^2 > 0$  we have*

$$\Pr[H - \mathbb{E}H \geq t] \leq \exp\left(-\frac{2t^2}{\sum_g c_g^2}\right).$$

If  $\sum_g c_g^2 = 0$ ,  $H$  is constant and its upper-tail probability is zero for  $t > 0$ .

*Proof.* This is the finite-valued bounded-differences inequality of McDiarmid [16], sufficient for every application here. We include its martingale proof and the bounded-range exponential-moment estimate [17].

Set  $M_g = \mathbb{E}[H \mid X_1, \dots, X_g]$ , with  $M_0 = \mathbb{E}H$ , and let  $Y_g = M_g - M_{g-1}$ . For a fixed past  $x_1, \dots, x_{g-1}$  of positive probability, independence makes

$$\phi_g(a) = \mathbb{E}_{X_{g+1}, \dots, X_M} H(x_1, \dots, x_{g-1}, a, X_{g+1}, \dots, X_M)$$

a function with  $|\phi_g(a) - \phi_g(b)| \leq c_g$ . Conditioned on that past,  $Y_g$  is  $\phi_g(X_g) - \mathbb{E}_{X_g} \phi_g(X_g)$ , so it has mean zero and lies in an interval of length at most  $c_g$ , not merely in an interval of length  $2c_g$ .

For completeness, if a finite-valued mean-zero random variable  $Y$  lies in an interval of length  $c$ , put  $k(s) = \log \mathbb{E} e^{sY}$ . Differentiating this finite sum shows that  $k''(s)$  is the variance under the probability law proportional to  $e^{sY}$  times the original law. Write  $\mathbb{E}_s$  for expectation under that law, which has the same range. With  $a$  the midpoint of a containing interval of length  $c$ , its variance is at most  $\mathbb{E}_s[(Y - a)^2] \leq c^2/4$ . Since  $k(0) = k'(0) = 0$ , integration twice gives  $k(s) \leq s^2 c^2/8$  for  $s \geq 0$ . Applying this to each conditional distribution yields

$$\mathbb{E}[e^{sY_g} \mid X_1, \dots, X_{g-1}] \leq e^{s^2 c_g^2/8}.$$

As  $\sum_g Y_g = H - \mathbb{E}H$ , iteration and Markov's inequality give, for  $s > 0$ ,

$$\Pr[H - \mathbb{E}H \geq t] \leq \exp\left(-st + \frac{s^2}{8} \sum_g c_g^2\right).$$

For  $t > 0$  and  $\sum_g c_g^2 > 0$ , minimize at  $s = 4t/\sum_g c_g^2$ . For  $t = 0$  the claimed bound is the trivial probability bound one. If all  $c_g = 0$ , changing coordinates one at a time shows that  $H$  is constant on its domain, proving the last claim.  $\square$

## 5 Exponential decay by expanding heavy gates

We strengthen the two-layer estimate to exponential decay in the number of one-input sources. All Hilbert spaces in this section are finite-dimensional. The resource is tensor-independent of the input sources, but it may now be represented by an arbitrary operator  $\sigma$  with  $\|\sigma\|_1 \leq 1$ , not necessarily Hermitian or positive. This extension is useful for subsequent exact expansions. Event factors remain positive contractions.

We repeatedly use partial-trace contractivity, including for non-Hermitian operators. If  $X$  acts on registers  $B, C$ , trace-norm duality gives

$$\|\mathrm{Tr}_B X\|_1 = \sup_{\|R\|_\infty \leq 1} |\mathrm{Tr}[(I_B \otimes R)X]| \leq \|X\|_1,$$

where  $R$  acts on  $C$ . Combining this with the trace-norm ideal inequality gives, for any operator  $T$  on  $B$ ,

$$\|\mathrm{Tr}_B[(T \otimes I)X]\|_1 \leq \|T\|_\infty \|X\|_1. \quad (12)$$

All inserted and partially traced operators below remain unnormalized.

**Lemma 8** (Mass and trace after rank-one insertion). *Let  $F$  be Hermitian,  $\mathrm{Tr} F = 0$ , and  $\|F\|_1 = 1$ . If  $T$  is a contraction of rank at most one on a tensor factor  $B$  and  $K = \mathrm{Tr}_B[(T \otimes I)F]$ , then*

$$\|K\|_1 + |\mathrm{Tr} K| \leq \|T\|_\infty \leq 1.$$

*Neither  $T$  nor  $K$  is required to be Hermitian.*

*Proof.* The case  $T = 0$  is immediate. Otherwise extract its sole singular value  $s = \|T\|_\infty$  and first prove the assertion for  $T = |u\rangle\langle v|$  with  $u, v$  unit. Put  $t = \mathrm{Tr} T = \langle v|u\rangle$ . For a unit vector  $z$  on  $B$ , set  $P_z = |z\rangle\langle z|$  and  $\omega = \langle z|T|z\rangle$ . Orthogonal decomposition and Cauchy–Schwarz give

$$\begin{aligned} |\omega| + |\omega - t| &\leq \|P_z u\| \|P_z v\| + \|(I - P_z)u\| \|(I - P_z)v\| \\ &\leq 1. \end{aligned}$$

Since  $(1 + e^{i\theta})\omega - t = (\omega - t) + e^{i\theta}\omega$ , for every pair of real phases  $\phi, \theta$ ,

$$\left| \Re(e^{i\phi}(1 + e^{i\theta})\omega) - \Re(e^{i\phi}t) \right| \leq 1.$$

Let  $U$  be a unitary on the complementary register. In an eigenbasis of  $U$ , the Hermitian part

$$H = \Re(e^{i\phi}T \otimes (I + U)), \quad \Re A = (A + A^*)/2,$$

is a direct sum of the operators  $\Re[e^{i\phi}(1 + e^{i\theta})T]$ , for eigenvalues  $e^{i\theta}$  of  $U$ . The preceding inequality bounds every unit-vector quadratic form in every block in the *same* interval  $[c - 1, c + 1]$ , where  $c = \Re(e^{i\phi}t)$  is independent of  $\theta$ . Hence  $\|H - cI\|_\infty \leq 1$ . Hermiticity and trace zero of  $F$  now give

$$\Re \mathrm{Tr}[(e^{i\phi}T \otimes (I + U))F] = \mathrm{Tr}(HF) = \mathrm{Tr}((H - cI)F), \quad |\mathrm{Tr}((H - cI)F)| \leq 1.$$

Choose a unitary  $U_0$  with  $\Re \mathrm{Tr}(U_0 K) = \|K\|_1$  by polar decomposition, extending the polar partial isometry if necessary. Choose a phase  $\eta$  with  $\Re(e^{i\eta} \mathrm{Tr} K) = |\mathrm{Tr} K|$ ; if the trace is zero, any phase works. Since

$$U_0 + e^{i\eta}I = e^{i\eta}(I + e^{-i\eta}U_0),$$

apply the preceding bound with  $\phi = \eta$  and  $U = e^{-i\eta}U_0$ . The partial-trace identity then yields

$$\|K\|_1 + |\mathrm{Tr} K| = \Re \mathrm{Tr}[(T \otimes (U_0 + e^{i\eta}I))F] \leq 1.$$

Multiplying back by  $s$  proves the stated inequality.  $\square$

**Lemma 9** (Retention after several source-port insertions). *Let  $F$  be a one-input source operator from Lemma 2, with port set  $\mathcal{J}$  and original weights  $h_j$ . Select a set  $B \subseteq \mathcal{J}$  of ports and insert a contraction  $T$  of rank at most one on their joint register:*

$$K = \text{Tr}_B[(T \otimes I)F], \quad \nu = \|K\|_1, \quad \tau = \text{Tr } K.$$

*For  $B = \emptyset$  use the scalar insertion  $T = 1$ , so  $K = F$ . Put  $\mathcal{R} = \mathcal{J} \setminus B$ . Complements for partial traces of  $K$  are taken within these remaining ports, not within  $\mathcal{J}$ .*

*There is a function  $f : 2^{\mathcal{R}} \rightarrow [0, \nu^2]$  such that*

- (i)  $\|\text{Tr}_{\mathcal{R} \setminus A} K\|_1^2 \leq f(A) \leq \nu^2 \leq 1$  for every  $A \subseteq \mathcal{R}$ ;*
- (ii) under independent half-retention of the remaining ports,  $\mathbb{E}f \leq 3\nu/4 \leq 3/4$ ;*
- (iii) for every  $A \subseteq \mathcal{R}$  and  $j \in \mathcal{R}$ ,*

$$|f(A \triangle \{j\}) - f(A)| \leq 4h_j.$$

*If  $B$  contains the owner, one may take  $f \equiv \nu^2$ , and then  $\nu \leq 1/2$ . Otherwise, one may take  $f = \nu^2$  when the owner is retained and the exact squared retained norm when it is erased; moreover,*

$$\mathbb{E}f \leq \frac{3\nu^2 + |\tau|^2}{4}. \quad (13)$$

*The original weights are not recomputed after insertion. In particular,  $T$  may be a tensor product of non-Hermitian rank-one contractions on any number of selected ports.*

*Proof.* Equation (12) gives  $\nu \leq 1$ . If  $T = 0$ , take  $f = 0$  and all assertions are immediate. If  $B$  contains the owner, its averaged-state marginal has a pure data-zero block of weight  $1/2$  and an orthogonal data-one block of total weight  $1/2$ , as in (4). Its largest eigenvalue is exactly  $1/2$ , even when  $B$  includes other ports. Lemma 4, applied once to that whole union, gives  $\nu \leq 1/2$ . The constant majorant  $f = \nu^2$  proves (i) and (iii) by partial-trace contractivity, and satisfies  $\mathbb{E}f = \nu^2 \leq \nu/2 \leq 3\nu/4$ .

Suppose henceforth that the owner remains. Work before the source's final port-local unitaries, absorbing their restrictions on  $B$  into the insertion vectors. This preserves all retained norms and does not identify the two layers' reflection axes. Write  $T = |u\rangle\langle v|$  with  $\|u\|, \|v\| \leq 1$ . Let  $\beta$  be the initial product vector on the inserted ancillary wires and  $\gamma$  the initial product vector on all remaining ancillary wires, including those in the owner. Let  $\Pi_B, \Pi_R$  be the corresponding tensor products of the first-layer single-qubit trigger projectors. Define

$$P_B = \langle \beta | \Pi_B | \beta \rangle, \quad P_R = \langle \gamma | \Pi_R | \gamma \rangle, \quad P_{\text{all}} = P_B P_R.$$

The input data wire is in neither ancillary product.

If  $P_{\text{all}} = 0$ , the first-layer reflection acts as the identity on the original two-dimensional input subspace. After inserting  $T$ , the remaining operator is a scalar multiple of the original signed data operator tensored with a pure ancillary state. It has trace zero. Taking  $f = \nu^2$  when the owner is retained and  $f = 0$  otherwise gives  $\mathbb{E}f = \nu^2/2$ . This is at most both  $3\nu^2/4$  and  $3\nu/4$ . Ancillary retention changes  $f$  by zero, and an owner toggle changes it by at most  $\nu^2 \leq 1 = 2h_0 \leq 4h_0$ . Thus every assertion holds in this case.

Assume  $P_{\text{all}} > 0$ , so  $P_B, P_R > 0$ . Choose a unit remaining trigger vector  $\eta$  with phase fixed by  $\Pi_R \gamma = \sqrt{P_R} \eta$ . For  $z = u, v$ , put

$$t_z = \langle z | \beta \rangle, \quad b_z = \langle z | \Pi_B \beta \rangle.$$

Cauchy–Schwarz implies

$$|t_z| \leq 1, \quad |b_z| \leq \sqrt{P_B}. \quad (14)$$

Define the compressed maps  $W_z$  from the data qubit into the remaining source register by their columns:

$$W_z|0\rangle = |0\rangle \otimes t_z\gamma, \quad W_z|1\rangle = |1\rangle \otimes (t_z\gamma - 2b_z\sqrt{P_R}\eta).$$

These maps are obtained by projecting the inserted ancillary factor of the reflection embedding  $a \mapsto G(a \otimes \beta \otimes \gamma)$  against  $z$ , where  $G$  is the first-layer reflection. They need not themselves be isometries. The signed data operator in the first gate's basis is

$$\frac{S}{2}, \quad S = \begin{pmatrix} \xi & \zeta \\ \bar{\zeta} & -\xi \end{pmatrix}, \quad \xi \in \mathbb{R}, \quad \xi^2 + |\zeta|^2 = 1.$$

Indeed, it is half the difference of the two orthogonal input projectors; with the convention in Lemma 2's proof,  $\xi = -\delta$ . The insertion order  $|u\rangle\langle v|$  gives exactly

$$K = \frac{1}{2}W_vSW_u^*. \quad (15)$$

This is a rank-at-most-two normal form, not a normalized source state. For empty insertion, the scalar conventions give  $P_B = t_u = t_v = b_u = b_v = 1$ .

We now compute a more general marginal: erase the data qubit and any subset of the remaining ancillary wires. Let  $C$  be the retained ancillary register, let  $p$  be its initial trigger probability, and let  $Q$  be the product of the erased ancillary trigger probabilities. Thus  $P_R = pQ$  and  $P_R \leq Q \leq 1$ . Write  $\rho_C = |\gamma_C\rangle\langle\gamma_C|$  for the retained initial product state and  $\Pi_C$  for its trigger projector. Put

$$x = t_v\bar{b}_u, \quad y = b_v\bar{t}_u, \quad z = b_v\bar{b}_u, \quad \tau_0 = P_R(x + y - 2z), \quad L = |x| + |y|.$$

The off-diagonal data blocks of (15) vanish. Expanding its two diagonal blocks and tracing the erased ancillas leaves the operator

$$K_C = \xi Q(x\rho_C\Pi_C + y\Pi_C\rho_C - 2z\Pi_C\rho_C\Pi_C).$$

For  $0 < p < 1$ , define the orthonormal vectors

$$\eta_C = \frac{\Pi_C\gamma_C}{\sqrt{p}}, \quad \omega_C = \frac{\gamma_C - \sqrt{p}\eta_C}{\sqrt{1-p}}.$$

The operator  $K_C$  acts within  $\text{span}\{\eta_C, \omega_C\}$  and vanishes on its orthogonal complement. In this ordered basis its matrix is

$$\xi \begin{pmatrix} \tau_0 & y\sqrt{P_R(Q - P_R)} \\ x\sqrt{P_R(Q - P_R)} & 0 \end{pmatrix}.$$

Its trace is  $\tau = \xi\tau_0$ , since partial trace preserves the full trace. For any 2-by-2 matrix  $M$  with singular values  $s_1, s_2$ ,

$$\|M\|_1^2 = (s_1 + s_2)^2 = \text{Tr}(M^*M) + 2|\det M|.$$

Consequently the exact squared norm is

$$\|K_C\|_1^2 = |\tau|^2 + \xi^2 L^2 P_R(Q - P_R). \quad (16)$$

When  $p = 1$ , the division-free expression for  $K_C$  is  $\xi\tau_0\rho_C$  and  $Q = P_R$ , so the same norm formula holds. This includes an empty retained register, when  $K_C$  is the scalar  $\tau$ . The case  $p = 0$  cannot occur in the present branch  $P_{\text{all}} > 0$ ; zero total overlap was handled separately above. Thus no continuity argument or division by zero is needed.

For actual port retention  $A \subseteq \mathcal{R}$ , define  $f(A) = \nu^2$  if the owner is retained. Otherwise define  $f(A) = D(A)^2$ , where  $D(A) = \|\text{Tr}_{\mathcal{R} \setminus A} K\|_1$  is given by (16). In this case all owner ancillas are erased along with the data. Partial-trace contractivity proves (i).

Let  $p_0$  be the trigger product on the owner ancillas and let  $p_j$  be the products on the remaining ancillary-only ports. Then  $P_R = p_0 \prod_j p_j$ . Conditioned on owner erasure,

$$\mathbb{E}Q = p_0 \prod_j \frac{1 + p_j}{2} \leq \frac{p_0 + P_R}{2} \leq \frac{1 + P_R}{2}.$$

The first inequality is the product inequality proved in Lemma 2; it also holds for an empty collection. Equation (16) gives

$$\mathbb{E}[D^2 \mid \text{owner erased}] \leq |\tau|^2 + \frac{1}{2}\xi^2 L^2 P_R(1 - P_R).$$

As an auxiliary partial trace, erase only the data and retain *all* remaining ancillas, even those sharing its owner port. Its trace norm is at most  $\nu$ . Setting  $Q = 1$  in the same formula yields

$$|\tau|^2 + \xi^2 L^2 P_R(1 - P_R) \leq \nu^2.$$

Therefore the conditional mean is at most  $(\nu^2 + |\tau|^2)/2$ . Averaging the independent owner coin proves

$$\mathbb{E}f \leq \frac{\nu^2}{2} + \frac{\nu^2 + |\tau|^2}{4} = \frac{3\nu^2 + |\tau|^2}{4},$$

which is (13).

Lemma 8 applied to the original  $F$  gives  $\nu + |\tau| \leq 1$ , also for the scalar empty insertion. If  $\nu \leq 1/2$ , the pointwise bound  $f \leq \nu^2$  gives (ii). For  $1/2 \leq \nu \leq 1$ ,

$$\mathbb{E}f \leq \frac{3\nu^2 + (1 - \nu)^2}{4} = \nu^2 - \frac{\nu}{2} + \frac{1}{4} \leq \frac{3\nu}{4},$$

because the difference from  $3\nu/4$  is  $(\nu - 1)(\nu - 1/4) \leq 0$ .

For (iii), suppose the owner is erased and ancillary port  $j$  changes from erased to retained. Writing  $Q_{\text{rest}} \leq 1$  for the product over the other erased ancillas,  $Q$  increases from  $p_j Q_{\text{rest}}$  to  $Q_{\text{rest}}$ . Hence the increase in  $f$  is

$$\xi^2 L^2 P_R Q_{\text{rest}}(1 - p_j) \leq 4P_B P_R(1 - p_j) = 4P_{\text{all}}(1 - p_j) \leq 4h_j.$$

Here (14) gives  $L \leq 2\sqrt{P_B}$ , and the last step is the *original* source bound (6). If the owner is retained, an ancillary toggle leaves  $f = \nu^2$  unchanged. Retaining the owner increases  $f$  by a number in  $[0, \nu^2]$ , hence by at most  $1 = 2h_0 \leq 4h_0$ . Erasing a retained port reverses the corresponding change, with the same absolute bound. This proves (iii).  $\square$

**Theorem 10** (Exponential two-layer source bound). *There are absolute constants  $K_0 \geq 1$  and  $a > 0$  with the following property. Let  $r \geq 0$  be an integer. Consider  $r$  independent one-input first-layer sources as in Section 3 and a tensor-independent resource operator  $\sigma$  with  $\|\sigma\|_1 \leq 1$ . Apply a second layer of pairwise wire-disjoint generalized controlled-Z gates, with arbitrary one-qubit gates before*

and after it. Use the nonempty source-by-second-gate ports and private idle ports of Theorem 5, including ports on the resource.

Let  $E$  be a tensor product of positive contractions over these ports. After absorbing the adjacent one-qubit gates as in that theorem, set

$$q(x) = \text{Tr} \left[ EU_2 \left( \sigma \otimes \bigotimes_{i=1}^r \rho_{i,x_i} \right) U_2^* \right], \quad x \in \{0,1\}^r,$$

where  $\rho_{i,x_i}$  is the state of input source  $i$  and  $U_2$  is the second multiqubit layer. Its full signed coefficient satisfies

$$|\hat{q}([r])| \leq K_0 e^{-ar}.$$

The functional  $q(x)$  is an event probability if  $\sigma$  is a density operator, and may be complex otherwise. The bound is independent of gate count, port widths, and finite resource dimension. For  $r = 0$  the coefficient is the single trace pairing, of magnitude at most one. An absent resource is the scalar 1.

*Proof.* The  $r = 0$  assertion follows from trace-norm duality. Assume  $r \geq 1$ . The signed operator before layer two is  $\Sigma = \sigma \otimes \bigotimes_{i=1}^r F_i$ . Use the original port weights  $h_{ig}$ , with  $\sum_g h_{ig} \leq C_s$  and  $h_{ig} = 0$  for absent ports. All rows refer only to input sources, not the resource. Fix a threshold  $w \geq 1$  for now. Call a real gate heavy if its input-source load  $\sum_i h_{ig}$  exceeds  $w$ . Let  $\mathcal{H}$  be the set of heavy gates and  $H = |\mathcal{H}|$ . The row budgets imply

$$H \leq C_s r / w. \tag{17}$$

**Expand all heavy gates.** For every  $g \in \mathcal{H}$ , expand its pulled-back event exactly:

$$D_g P_g D_g = P_g - 2\Pi_g P_g - 2P_g \Pi_g + 4\Pi_g P_g \Pi_g.$$

After removing the scalar coefficients  $-2, -2, 4$ , the three correction operators factor over source ports into contractions of rank at most one, as in (9). A zero correction can be omitted. The base term  $P_g$  is a product of positive port effects and is split into private vertices.

Choose a correction set  $S \subseteq \mathcal{H}$  and, for each  $g \in S$ , a type  $\ell_g \in \{1, 2, 3\}$ . Denote its *coefficient-free* scalar term by  $\mathcal{T}_{S,\ell}$ . With  $(a_1, a_2, a_3) = (-2, -2, 4)$ ,

$$\hat{q}([r]) = \sum_{S \subseteq \mathcal{H}} \sum_{\ell \in \{1,2,3\}^S} \left( \prod_{g \in S} a_{\ell_g} \right) \mathcal{T}_{S,\ell}.$$

For  $k = |S|$ , the absolute coefficient mass of the inner sum is at most  $(2 + 2 + 4)^k = 8^k$ , with equality before zero terms are omitted. The empty correction set has one term with coefficient one.

Fix  $(S, \ell)$ . On source  $i$ , the corrections form a contraction  $T_i$  of rank at most one on the union  $J_i$  of its selected ports. Insert it once on that whole union and trace:

$$K_i = \text{Tr}_{J_i} [(T_i \otimes I) F_i], \quad \nu_i = \|K_i\|_1.$$

For  $J_i = \emptyset$  take the scalar  $T_i = 1$ , so  $K_i = F_i$ . The resource remainder has norm at most one by (12). No operator is renormalized. The remaining real gates are light; unselected heavy gates contribute private positive effects. These surviving effects act on wires disjoint from the inserted ports and remain positive contractions. All remaining ports keep their original weights.

**Uniform bound for every surviving light network.** Let  $\mathcal{V}$  be the surviving real/private vertex set and  $\mathcal{V}_i$  the subset incident to the remaining ports of source  $i$ . Use the majorants of Lemma 9, with these ports identified with  $\mathcal{V}_i$ . Expanding each effect as  $(I + A_g)/2$  using independent fair vertex coins gives, as in (10),

$$|\mathcal{T}_{S,\ell}| \leq \mathbb{E}_A \prod_i D_i(A), \quad D_i(A) = \|\text{Tr}_{\mathcal{V}_i \setminus A} K_i\|_1.$$

The resource remainder may be non-Hermitian, but its further partial traces have norm at most one. No factorization across its ports is used. Different sources may share a coin, while the distinct ports of any one source have distinct independent coins.

Put  $F_*(A) = \sum_i f_i(A \cap \mathcal{V}_i)$ . Its mean is at most  $3r/4$ . Changing vertex  $g$  changes it by at most  $c_g = 4 \sum_i h_{ig}$ . Remaining row sums are at most  $C_s$ , real column loads are at most  $w$ , and private column loads are at most  $1/2$ . Consequently

$$\sum_g c_g^2 \leq 16C_s r w.$$

If this squared sum is positive, Lemma 7 gives  $\Pr[F_* \geq 7r/8] \leq e^{-r/(512C_s w)}$ . If it is zero,  $F_*$  is constant with value at most  $3r/4$ , so that exceptional event is empty. On the complementary event AM–GM gives  $\prod_i D_i \leq (F_*/r)^{r/2} \leq (7/8)^{r/2}$ ; on the exceptional event every  $D_i \leq 1$ . Thus

$$|\mathcal{T}_{S,\ell}| \leq \left(\frac{7}{8}\right)^{r/2} + e^{-r/(512C_s w)} \leq 2e^{-c_0 r/w}, \quad c_0 = \frac{1}{512C_s}. \quad (18)$$

The last step uses  $w \geq 1$  and  $\frac{1}{2} \log(8/7) > c_0$ :  $\log(8/7) \geq 1/8$  and  $C_s > 1/2$  already suffice. This bound is uniform in  $S$ , in its correction types, and in the number of surviving vertices, including an empty vertex set.

**A joint penalty for many corrections.** For every source  $i$ , let  $P_i$  be the product of the original first-layer trigger probabilities on all its ancillas, including those in the owner. These are first-layer probabilities; no alignment with the second-layer reflection axes is assumed. Define

$$W_i(S) = \sum_{g \in S} h_{ig}, \quad H_i(S) = 1 - \lambda_{\max}(\bar{\rho}_{i,J_i}),$$

where  $\bar{\rho}_{i,J_i}$  is the joint marginal of the *original* averaged source state on  $J_i$ . For an empty union it is the scalar 1, so  $H_i(S) = W_i(S) = 0$ . Lemma 4, applied once to the whole union, gives  $\nu_i \leq 1 - H_i(S)$ . If  $J_i$  contains the owner, the data-block argument from (4) gives

$$H_i(S) = \frac{1}{2} \geq \frac{W_i(S)}{2C_s}. \quad (19)$$

For a nonempty ancillary-only union, let  $p_j$  be the product on selected port  $j$ , and put  $p_{J_i} = \prod_{j \in J_i} p_j$ . If  $P_i = 0$ , all ancillary-only ports and their union remain pure, so  $W_i(S) = H_i(S) = 0$ , without using logarithms. For  $P_i > 0$ , one has  $p_{J_i} \geq P_i > 0$ . Applying the calculation (5)–(6) to the union as one port gives

$$H_i(S) \geq P_i(1 - p_{J_i}), \quad W_i(S) \leq 2P_i \sum_{j \in J_i} (1 - p_j) \leq 2P_i(-\log p_{J_i}). \quad (20)$$

Fix  $\delta \in (0, e^{-1})$ . If  $P_i \geq \delta$  and  $p_{J_i} < 1$ , the increasing function  $(1 - p)/(-\log p)$  gives

$$H_i(S) \geq \frac{1 - \delta}{2 \log(1/\delta)} W_i(S).$$

Its derivative has the sign of  $\log p + p^{-1} - 1$ , which is nonnegative: the latter function decreases to zero as  $p$  increases to one. If  $p_{J_i} = 1$  or  $J_i$  is empty, the same bound holds because  $W_i(S) = 0$ .

If  $0 < P_i < \delta$  and the owner is not selected, its entire ancillary weight is at most

$$2P_i \log(1/P_i) \leq 2\delta \log(1/\delta),$$

because  $P \log(1/P)$  is increasing on  $(0, e^{-1})$ . For  $P_i = 0$  that weight is zero directly. Owner-containing unions always retain (19), even for small or zero  $P_i$ ; those weights are not discarded. Set

$$c_\delta = \min \left\{ \frac{1}{2C_s}, \frac{1 - \delta}{2 \log(1/\delta)} \right\} > 0.$$

For  $k = |S|$ , heaviness gives  $\sum_i W_i(S) \geq kw$ , with strictness when  $k > 0$ . Discarding only the owner-free, small- $P_i$  rows loses at most  $2r\delta \log(1/\delta)$ , so

$$\sum_i H_i(S) \geq c_\delta(kw - 2r\delta \log(1/\delta)).$$

This also holds for  $S = \emptyset$  and for a negative bracket. The remaining event is a contraction, so

$$|\mathcal{T}_{S,\ell}| \leq \prod_i \nu_i \leq e^{-\sum_i H_i(S)} \leq \exp[-c_\delta(kw - 2r\delta \log(1/\delta))]. \quad (21)$$

Only one joint eigenvalue penalty per source is used, never a product of its individual-port penalties.

**Sparse and dense correction sets.** Choose a fixed  $\varepsilon_0 \in (0, C_s)$  such that

$$A_0 := \varepsilon_0 \log(8eC_s/\varepsilon_0) < c_0.$$

Such a choice exists because  $\varepsilon \log(1/\varepsilon) \rightarrow 0$  as  $\varepsilon \downarrow 0$ . Next choose a fixed  $\delta \in (0, e^{-1})$  with

$$2\delta \log(1/\delta) \leq \varepsilon_0/2.$$

Finally choose a fixed  $w \geq 1$  large enough that

$$\frac{8eC_s}{\varepsilon_0} e^{-c_\delta w/2} \leq e^{-1}. \quad (22)$$

This is possible since  $c_\delta > 0$  after  $\delta$  is fixed. All choices are absolute and independent of the circuit. Put  $u = r/w > 0$  and  $b = c_0 - A_0 > 0$ , so  $H \leq C_s u$ .

For integers  $1 \leq k \leq \min\{H, \varepsilon_0 u\}$ ,

$$\binom{H}{k} 8^k \leq \exp(k \log(8eC_s u/k)) \leq e^{A_0 u}.$$

The first step uses  $\binom{H}{k} \leq (eH/k)^k$ . For the second, the derivative of  $s \log(8eC_s u/s)$  is  $\log(8C_s u/s) > 0$  on  $0 < s \leq \varepsilon_0 u$ , since  $\varepsilon_0 < C_s$ . The  $k = 0$  summand is one and is also at most  $e^{A_0 u}$ . There are at most  $H + 1 \leq C_s u + 1$  summands; hence

$$\sum_{\substack{0 \leq k \leq H \\ k \leq \varepsilon_0 u}} \binom{H}{k} 8^k \leq (C_s u + 1) e^{A_0 u}.$$

Multiplying by the uniform bound (18) shows that all sparse correction terms contribute at most  $2(C_s u + 1)e^{-bu}$ .

For integers  $\varepsilon_0 u < k \leq H$ ,

$$2r\delta \log(1/\delta) \leq \frac{\varepsilon_0 r}{2} < \frac{kw}{2}.$$

The bracket in (21) is therefore at least  $kw/2$ . Using that bound for the dense correction terms gives

$$\begin{aligned} \binom{H}{k} 8^k e^{-c_\delta kw/2} &\leq \left( \frac{8eC_s}{\varepsilon_0} e^{-c_\delta w/2} \right)^k \\ &\leq e^{-k}. \end{aligned}$$

Extending the sum to all integers  $k > \varepsilon_0 u$  bounds it by

$$\frac{e^{-(\lfloor \varepsilon_0 u \rfloor + 1)}}{1 - e^{-1}} \leq \frac{e^{-\varepsilon_0 u}}{1 - e^{-1}}.$$

These estimates cover  $H = 0$  and  $\varepsilon_0 u < 1$ : empty dense sums contribute zero and the sparse sum still includes  $k = 0$ .

Combining both contributions proves

$$|\hat{q}([r])| \leq 2(C_s r/w + 1)e^{-(c_0 - A_0)r/w} + \frac{e^{-\varepsilon_0 r/w}}{1 - e^{-1}}. \quad (23)$$

To make the final absorption explicit, use  $\sup_{u \geq 0} ue^{-bu/2} = 2/(eb)$  and choose

$$a = \frac{\min\{b/2, \varepsilon_0\}}{w} > 0, \quad K_0 = 2 + \frac{4C_s}{eb} + \frac{1}{1 - e^{-1}} > 1.$$

Then  $2(C_s u + 1)e^{-bu} \leq (2 + 4C_s/(eb))e^{-bu/2}$ , and each exponential on the right of (23) is bounded by  $e^{-ar}$ . This proves  $|\hat{q}([r])| \leq K_0 e^{-ar}$  for  $r \geq 1$ ; the same  $K_0 > 1$  includes the already verified case  $r = 0$ .  $\square$

## 6 Exact first-layer cleanup

We use the parity-preserving quantum restriction of [15, Lemma 4.1]. Its preparation of the fixed two-qubit states within the original first layer is also part of that proof. We give the parity-only argument with explicit circuit and resource accounting. Here a *restriction* may fix a pair of input wires to a superposition of computational-basis strings of the same parity; it is not necessarily an ordinary classical bit restriction.

**Lemma 11** (Parity-preserving cleanup). *Let  $C$  be an exact  $n$ -input parity circuit in the model of Section 2, on  $N$  physical wires, with  $n \geq 1$ . There is a circuit  $C'$  on the same wires, with  $r \geq \lceil n/3 \rceil$  variable inputs, that computes parity or its complement exactly on the same designated output wire and does not increase entangling depth or multiqubit-gate count. Each first-layer component of  $C'$  contains at most one variable input. Components here are gate supports and individual idle wires, including untouched input singletons. All  $N - r$  ancillary wires of  $C'$  are initialized in a fixed pure product state. Their number may exceed the original  $N - n$ : it is physical wire count, not ancillary wire count, that is preserved.*

*Proof.* If the circuit has no entangling layer, leave it unchanged and take  $r = n$ ; its components are singletons. Otherwise fix its first entangling layer, allowing that layer to be empty.

**Exactness permits fixed-parity substitutions.** Use the output eigenvectors  $|a_{\pm}\rangle$  from Section 2. For the original product ancillary state  $|\alpha\rangle$ , exactness gives

$$C(|x\rangle \otimes |\alpha\rangle) = |a_{\chi(x)}\rangle * o \otimes |\gamma_x\rangle.$$

Let  $\mathcal{I} \subseteq [n]$  be a set of retained input positions. For a string  $u$  on any indicated subset  $S \subseteq [n]$ , use  $\chi(u) = (-1)^{\sum_{j \in S} u_j}$ ; in particular the empty string has parity sign one. Suppose the other input positions are fixed to a unit vector  $\Phi = \sum_t c_t |t\rangle$  supported on strings with  $\chi(t) = \tau \in \{\pm 1\}$ . All tensor factors are placed on their original physical wires. For  $z \in \{0, 1\}^{\mathcal{I}}$ , let  $x(z, t)$  combine the retained bits  $z$  and fixed-position bits  $t$ , and set

$$J|z\rangle = \sum_t c_t |x(z, t)\rangle \otimes |\alpha\rangle.$$

This map is an isometry. Since  $\chi(x(z, t)) = \chi(z)\chi(t) = \tau\chi(z)$  on the support of  $\Phi$ , linearity gives

$$CJ|z\rangle = |a * \tau\chi(z)\rangle * o \otimes \left( \sum_t c_t |\gamma * x(z, t)\rangle \right).$$

The parenthesized garbage vector has norm one because  $CJ$  is an isometry and the output factor is a unit vector. In particular, interference among the garbage vectors cannot spoil the deterministic output. Thus such a substitution preserves parity up to the fixed sign  $\tau$ . The proof uses exactness, not merely a bounded-error guarantee on computational-basis inputs.

**Deactivate gates without rotating the input parity basis.** Write  $C = C_{>1} D_1 L_0$ , where  $L_0$  is the initial product of one-qubit unitaries. Set

$$R_1 = L_0^* D_1 L_0, \quad V = C_{>1} L_0, \quad C = V R_1.$$

The gates of  $R_1$  are disjoint product reflections with the same supports as those of  $D_1$ . The factor  $L_0$  has moved to the following local layer and can be merged with it; the computational input basis and the parity classes above have not been changed.

Consider a gate  $G_g = I - 2|\theta_g\rangle\langle\theta_g|$  of  $R_1$ , where  $|\theta_g\rangle$  is a product vector, and suppose its support contains  $k \geq 3$  original input wires. Choose two, denoted  $i, j$ , and write their trigger factor as  $|\theta_i\theta_j\rangle$ . Put

$$a = \langle\theta_i\theta_j|01\rangle, \quad b = \langle\theta_i\theta_j|10\rangle.$$

Choose a unit vector  $\phi_g = \alpha_g|01\rangle + \beta_g|10\rangle$  by

$$(\alpha_g, \beta_g) = \begin{cases} (b, -a)/\sqrt{|a|^2 + |b|^2}, & (a, b) \neq (0, 0), \\ (1, 0), & (a, b) = (0, 0). \end{cases}$$

Then  $a\alpha_g + b\beta_g = 0$ , including for complex  $a, b$ , and  $\phi_g$  has odd computational parity. The tensor-product form of the trigger projector implies

$$(|\theta_g\rangle\langle\theta_g|)(\phi_g \otimes \psi) = 0, \quad G_g(\phi_g \otimes \psi) = \phi_g \otimes \psi$$

for every vector  $\psi$  on the other wires in the gate. Thus fixing this pair to  $\phi_g$  deactivates the entire gate, while all other  $k - 2$  input wires remain variable. The selected pairs are disjoint because the original gate supports are disjoint.

For a first-layer gate with exactly two input wires, fix one of them to  $|0\rangle$  and retain the gate on its original support. It then has only one variable input; the fixed wire becomes an ancilla. Leave gates with one or zero input wires unchanged, and leave every idle input wire variable. In particular, an input-free gate contributes nothing to the count of variable inputs.

**Prepare the pairs from product initialization in the same layer.** For each deactivated gate define

$$|\omega_g\rangle = (\alpha_g|0\rangle + \beta_g|1\rangle) \otimes |1\rangle, \quad P_g = \text{CNOT}_{i \rightarrow j}.$$

The direct identity

$$P_g|\omega_g\rangle = \alpha_g|01\rangle + \beta_g|10\rangle = \phi_g$$

prepares the required pair from a product state. Moreover  $P_g = H_j \text{CZ}_{\{i,j\}} H_j$ , so it costs one allowed multiqubit gate and only free one-qubit unitaries.

Let  $\mathcal{I}$  now be the surviving input positions. Let  $P$  be the product of these pair preparations, and let  $R_{\text{keep}}$  be the product of the first-layer reflections that were not deactivated. Their supports are mutually disjoint and are disjoint from the prepared pairs. Let  $|\Omega_z\rangle$  be the product input state with retained bits  $z$ , each selected pair in  $\omega_g$ , each classically fixed input in  $|0\rangle$ , and all original ancillas still in  $\alpha$ . If  $\Phi$  is the tensor product of the  $\phi_g$  and the classically fixed zero bits, then

$$P|\Omega_z\rangle = J|z\rangle, \quad R_1 J|z\rangle = R_{\text{keep}} J|z\rangle.$$

Consequently the circuit

$$C' = V R_{\text{keep}} P \quad \text{satisfies} \quad C'|\Omega_z\rangle = C J|z\rangle$$

for every  $z$ . The state  $\Phi$  has fixed parity  $\tau = (-1)^t$ , where  $t$  is the number of prepared pairs, so the preceding exactness argument applies. If  $\tau = -1$ , interchanging the two output labels gives parity itself without adding any gate.

The layer  $R_{\text{keep}} P$  replaces each deactivated gate by one two-qubit gate on a subset of its original support and retains all other gates. It is therefore one entangling layer, with no increase in multiqubit-gate count. The local conjugations that implement its product reflections and CNOTs can be merged into the adjacent local layers. The suffix  $V$  uses the original later multiqubit layers, so the full entangling depth does not increase. All physical wires, including the output wire even if it was one of the fixed inputs, are retained.

Each prepared pair is now an input-free first-layer component. All other wires of a deactivated gate are singleton components; in particular its surviving variable inputs are distinct sources. The undeactivated input-containing gates have at most one variable input each. Since  $\Omega_z$  uses only fixed single-qubit ancillary states, initialization remains a pure product. This does not assert that the ancillary state after the first layer is product.

**Count the retained inputs.** Partition the original wires into first-layer gate supports and idle singletons. A component containing  $k$  input wires retains

$$t(k) = \begin{cases} 0, & k = 0, \\ 1, & k = 1 \text{ or } k = 2, \\ k - 2, & k \geq 3. \end{cases}$$

For every integer  $k \geq 0$ ,  $t(k) \geq k/3$ ; equality in the nontrivial  $k \geq 3$  case occurs at  $k = 3$ . Since the component input counts sum to  $n$ , their retained counts sum to an integer  $r \geq n/3$ , hence  $r \geq \lceil n/3 \rceil$ . This also covers  $n = 1, 2$  and arbitrary input-free or idle components.  $\square$

## 7 Small-support observables after three layers

We convert a product of one-qubit test operators at the output of three layers into a linear combination of product events after two layers. The expansion is algebraic; the cleaned first-layer assumption is needed only when applying the event bounds.

**Lemma 12** (Small-support pullback). *Consider a three-layer prefix  $C^{(3)}$  in the model of Section 2, on  $N \geq r \geq 1$  qubits with variable input  $x \in \{0,1\}^r$  and fixed pure product ancillas. Empty multiqubit layers and arbitrary intervening and boundary one-qubit gates are allowed. Suppose every input-containing first-layer component, including an untouched input singleton, contains exactly one variable input. Let  $m \geq 0$  be the number of second-layer multiqubit gates.*

*Let  $s \geq 1$  be an integer and let  $S$  be a fixed set of at most  $s$  physical qubits. For each  $q \in S$ , let  $B_q$  be a fixed one-qubit contraction, independent of  $x$ , and put*

$$B = \left( \bigotimes_{q \in S} B_q \right) \otimes I_{S^c}.$$

*Write  $\langle B \rangle_x = \text{Tr}(\rho_x^{(3)} B)$ , where  $\rho_x^{(3)}$  is the state after the entire prefix, and put  $\chi(x) = (-1)^{x_1 + \dots + x_r}$ . Then, with uniform averages over  $x$ ,*

$$|\mathbb{E}_x \chi(x) \langle B \rangle_x| \leq 36^s \Delta_r(m, w) \quad \text{for every real } w \geq 1,$$

*where  $\Delta_r(m, w)$  is defined in (8). Also,*

$$|\mathbb{E}_x \chi(x) \langle B \rangle_x| \leq 36^s K_0 e^{-ar},$$

*with the absolute constants from Theorem 10. Neither Hermiticity of  $B$  nor exact parity computation by the prefix is assumed; the expectation may be complex.*

*Proof. Step 1: local gates and affected supports.* Write

$$C^{(3)} = L_3 D_3 C^{(2)}, \quad C^{(2)} = L_2 D_2 L_1 D_1 L_0,$$

where the  $L_j$  are products of one-qubit unitaries and the  $D_j$  are disjoint controlled- $Z$  layers. Let  $\rho_x^{(2)}$  be the output state of  $C^{(2)}$  and put  $\tilde{B} = L_3^* B L_3$ . Conjugation by  $L_3$  preserves the one-qubit tensor form, the support bound, and the contraction bounds. Cyclicity of trace gives

$$\langle B \rangle_x = \text{Tr} \left[ \rho_x^{(2)} D_3 \tilde{B} D_3 \right].$$

Set  $t = |S| \leq s$ . A third-layer gate disjoint from  $S$  conjugates identity and has no effect. If  $h$  gates meet  $S$ , then  $h \leq t$ , because the gate supports are pairwise disjoint. Supported qubits idle in this layer remain as individual tensor factors.

*Step 2: expansion of the affected reflections.* For an affected gate on  $T_g$ , write  $D_g = I - 2\Pi_g$  with  $\Pi_g = \bigotimes_{q \in T_g} \pi_q$ , where each  $\pi_q$  is a rank-one orthogonal projector. For the restriction  $\tilde{B}_g = \bigotimes_{q \in T_g} \tilde{B}_q$ , with  $\tilde{B}_q = I$  for  $q \notin S$ , the exact identity is

$$D_g \tilde{B}_g D_g = \tilde{B}_g - 2\Pi_g \tilde{B}_g - 2\tilde{B}_g \Pi_g + 4\Pi_g \tilde{B}_g \Pi_g.$$

After separating the scalar coefficients 1,  $-2$ ,  $-2$ , 4, the four branches have one-qubit factors

$$\tilde{B}_q, \quad \pi_q \tilde{B}_q, \quad \tilde{B}_q \pi_q, \quad \pi_q \tilde{B}_q \pi_q,$$

respectively. Each factor has operator norm at most one by submultiplicativity. When  $q \notin S$ , these factors are  $I$  in the base branch and  $\pi_q$  in each correction branch. Thus the simultaneous expansion over all  $h$  affected gates is a finite sum of physical-qubit product tensors with total coefficient absolute sum at most

$$(1 + 2 + 2 + 4)^h = 9^h.$$

The factors at the original  $t$  sites may be non-Hermitian contractions; every other factor is already an identity or a rank-one orthogonal projector. Zero terms may be omitted.

*Step 3: decomposition of the remaining qubit factors.* For any one-qubit contraction  $M$ , write

$$M = H + iK, \quad H = (M + M^*)/2, \quad K = (M - M^*)/(2i).$$

Both  $H$  and  $K$  are Hermitian contractions, since  $\|M^*\|_\infty = \|M\|_\infty \leq 1$  and the operator norm satisfies the triangle inequality. Choose orthonormal eigenbases separately to write

$$H = \sum_{a=1}^2 \lambda_a P_a, \quad K = \sum_{b=1}^2 \mu_b Q_b, \quad |\lambda_a|, |\mu_b| \leq 1,$$

where the  $P_a, Q_b$  are rank-one orthogonal projectors. Repeated eigenvalues cause no difficulty: choose an orthonormal basis of the eigenspace; zero coefficients may be discarded. Consequently  $M$  is a linear combination of such projectors with coefficient absolute sum at most four. This need not be a convex combination, and the coefficients for  $iK$  may be imaginary.

Apply this decomposition at each of the original  $t$  sites in each expanded tensor. Tensoring the decompositions multiplies their coefficient absolute sums, so

$$D_3 \tilde{B} D_3 = \sum_{\nu} c_{\nu} E_{\nu}, \quad \sum_{\nu} |c_{\nu}| \leq 9^h 4^t \leq 36^s,$$

where every  $E_{\nu}$  is a physical-qubit product event: each factor is an identity or a rank-one orthogonal projector. The events and coefficients are fixed independently of  $x$ . For  $t = 0$ , there is just the identity event with coefficient one; the same estimates hold using empty-product conventions. No first-layer hypothesis was used in this algebraic expansion.

*Step 4: application of the two-layer event bounds.* For  $q_{\nu}(x) = \text{Tr}(\rho_x^{(2)} E_{\nu})$ , absorb the final local layer  $L_2$  of  $C^{(2)}$  into  $E_{\nu}$ . This again preserves its physical-qubit product-event form. Grouping its one-qubit factors within each nonempty source-second-layer port produces a positive contraction on that port, exactly as required by Theorems 5 and 10.

The first-layer assumption supplies  $r$  independent one-input sources. All input-free first-layer components can be grouped into a density-operator resource tensor-independent of those sources; this follows from product initialization and first-layer disjointness. Thus every  $q_{\nu}$  satisfies both  $|\hat{q}_{\nu}([r])| \leq \Delta_r(m, w)$  and  $|\hat{q}_{\nu}([r])| \leq K_0 e^{-ar}$ . Finally, linearity of the uniform signed average gives

$$\mathbb{E}_x \chi(x) \langle B \rangle_x = \sum_{\nu} c_{\nu} \hat{q}_{\nu}([r]).$$

The triangle inequality and  $\sum_{\nu} |c_{\nu}| \leq 36^s$  prove both claims. If  $S$  is empty, the coefficient is in fact zero because  $r \geq 1$  and  $\langle I \rangle_x = 1$ .  $\square$

## 8 The exact last-layer certificate

The following argument uses only the final layer and exact output determinism, not the depth or internal structure of the prefix. We use the identity/active-sector decomposition from [15, Section 2.3.1] to derive an explicit quantitative certificate on one- and two-qubit product observables.

**Lemma 13** (Exact last-layer certificate). *Let a circuit in the model of Section 2 compute parity exactly on  $n \geq 1$  input bits. Suppose the final multiqubit layer contains a gate incident to the output qubit  $o$ , with  $b \geq 1$  other wires. Regard those wires as its controls in the product-reflection representation. Let  $\rho_x$  be the state at the output of the prefix preceding that layer, with any preceding local gates included, and put*

$$\alpha(B) = 2^{-n} \sum_{x \in \{0,1\}^n} \chi(x) \operatorname{Tr}(\rho_x B), \quad \chi(x) = (-1)^{x_1 + \dots + x_n}.$$

*There is a fixed Hermitian product contraction  $B$ , independent of  $x$  and supported on at most two physical prefix-output qubits, such that*

$$\alpha(B) \geq \frac{1}{2b+1}.$$

*Here a product contraction means a tensor product of one-qubit contractions, with identity on every unspecified wire. If no gate in the final multiqubit layer touches  $o$ , there is instead a one-qubit Hermitian unitary  $B$  with  $\alpha(B) = 1$ .*

*Proof. Step 1: isolate the output gate.* Absorb the terminal local rotation on  $o$  into its measurement observable  $A_o$ . Exactness and  $n \geq 1$  make  $A_o$  a Hermitian unitary, by Section 2. All other terminal local gates act outside  $o$  and do not change its statistics. Within the final multiqubit layer, gate supports are disjoint. Writing that layer as  $UG$ , where  $G$  is the gate incident to  $o$ , the other gates  $U$  commute with both  $G$  and  $A_o$ . Hence

$$(UG)^* A_o (UG) = G A_o G.$$

Thus removing  $U$  does not change the output expectation, even when  $\rho_x$  is entangled across their supports.

Label the control wires by  $1, \dots, b$ , separately from the input indices. In the gate's product basis, write

$$P = \bigotimes_{j=1}^b P_j, \quad Q = I - P, \quad R_o = I_o - 2\pi_o, \quad G = Q \otimes I_o + P \otimes R_o,$$

where  $P_j$  and  $\pi_o$  are one-qubit rank-one orthogonal projectors. This representation is valid with any incident wire chosen as the output; no distinguished target of a Toffoli representation is needed. Suppressing identity tensor factors, set  $T = G A_o G$ . For every input,  $y = \chi(x)$  satisfies

$$\operatorname{Tr}(\rho_x T) = y, \quad H_x := I - yT \geq 0, \quad \operatorname{Tr}(\rho_x H_x) = 0.$$

Indeed,  $T$  is a Hermitian unitary and  $\operatorname{Tr} \rho_x = 1$ .

*Step 2: exactness on each inactive control sector.* For  $Q_j = I - P_j$ , extended by identity off control  $j$ , we have  $Q_j P = 0$ ,  $Q_j G = G Q_j = Q_j$ , and  $[Q_j, A_o] = 0$ . Consequently  $Q_j$  commutes with both  $T$  and  $H_x$  and  $Q_j T = Q_j A_o$ . Since  $Q_j$  is an orthogonal projector,

$$0 \leq Q_j H_x \leq H_x.$$

Indeed, block decomposition with respect to  $Q_j$  writes  $H_x$  as the sum of the positive operators  $Q_j H_x Q_j$  and  $(I - Q_j) H_x (I - Q_j)$ , which proves this order inequality. Positivity of  $\rho_x$  and  $\text{Tr}(\rho_x H_x) = 0$  now give

$$0 = \text{Tr}(\rho_x Q_j H_x) = \text{Tr}(\rho_x Q_j) - y \text{Tr}(\rho_x A_o Q_j).$$

No projection of the state is performed or normalized here. Averaging the identity yields

$$p_j := \mathbb{E}_x \text{Tr}(\rho_x Q_j) = \alpha(A_o Q_j) \geq 0.$$

The commuting control projectors satisfy the operator union bound

$$Q = I - \prod_{j=1}^b (I - Q_j) \leq \sum_{j=1}^b Q_j.$$

To see this, simultaneously diagonalize the  $Q_j$ ; each eigenvalue is 0 or 1, and the indicator that at least one eigenvalue is 1 is at most the number of 1's. It follows that

$$q := \mathbb{E}_x \text{Tr}(\rho_x Q) \leq \sum_{j=1}^b p_j.$$

No statistical independence of the control outcomes is assumed.

*Step 3: obtain the quantitative certificate.* Put  $D = R_o A_o R_o$ , a one-qubit Hermitian unitary. The orthogonal  $P, Q$  sectors give the exact identities

$$T = Q A_o + P D, \quad T - D = Q(A_o - D).$$

Because  $A_o - D$  is Hermitian of norm at most two and acts only on  $o$ , it commutes with  $Q$ . Therefore

$$-2Q \leq T - D \leq 2Q, \quad |\text{Tr}(\rho_x(T - D))| \leq 2 \text{Tr}(\rho_x Q).$$

Since  $\alpha(T) = \mathbb{E}_x y^2 = 1$ , averaging with the parity sign gives

$$|1 - \alpha(D)| \leq 2q.$$

All the coefficients here are real. Keeping their signs and using the bound on  $q$ , we obtain

$$1 \leq \alpha(D) + 2q \leq \alpha(D) + 2 \sum_{j=1}^b \alpha(A_o Q_j).$$

In particular, the following absolute-value form will also be useful:

$$1 \leq |\alpha(R_o A_o R_o)| + 2 \sum_{j=1}^b |\alpha(A_o Q_j)|. \quad (24)$$

The right-hand side of the signed inequality is at most

$$(2b + 1) \max\{\alpha(D), \alpha(A_o Q_1), \dots, \alpha(A_o Q_b)\}.$$

Thus at least one of these coefficients is at least  $1/(2b + 1)$ , with its positive sign as claimed. The corresponding operator is either the one-qubit Hermitian unitary  $D$  or a product  $A_o Q_j$  of one-qubit Hermitian contractions on two distinct wires. All choices are fixed independently of the input.

If the final multiqubit layer has no gate on  $o$ , the same conjugation argument removes all its gates from the output expectation. Exactness then gives  $\text{Tr}(\rho_x A_o) = \chi(x)$  and hence  $\alpha(A_o) = 1$ . This also covers an empty final layer; a circuit with no multiqubit layer can be padded with an empty layer. The proof uses neither purity of  $\rho_x$  nor a cleaned first layer; its positivity argument does require the pointwise exactness assumption.  $\square$

## 9 Control cells: removing any ancilla-count assumption

The preceding certificate has a prefactor depending on the number of controls of the final gate. We replace individual controls by blocks, called *cells*, determined by their first-layer source and their second- and third-layer incidences. Each cell lies in a single two-layer port and meets at most one third-layer gate. These two properties allow its entire control projector to be treated as one rank-one operator, with a cost independent of the number of physical qubits in the cell.

**Lemma 14** (Cell certificate). *Consider an exact parity circuit in the model of Section 2, on  $N \geq r \geq 1$  qubits with  $r$  variable inputs and fixed pure product ancillas. Write it with four multiqubit layers, allowing empty layers and arbitrary intervening and boundary one-qubit gates. Suppose every input-containing first-layer component, including an idle input singleton, contains exactly one variable input. Let  $m_2, m_3 \geq 0$  be the numbers of multiqubit gates in layers two and three.*

*There is a nonnegative integer  $N_{\text{cell}}$ , depending only on the circuit, such that, for every real  $w \geq 1$ ,*

$$1 \leq [36 + 2664N_{\text{cell}}]\Delta_r(m_2, w), \quad N_{\text{cell}} \leq (r+1)(m_2+1)(m_3+1). \quad (25)$$

*Here  $\Delta_r$  is defined in (8). If the fourth multiqubit layer does not touch the output, one may take  $N_{\text{cell}} = 0$ . The same inequality holds with  $\Delta_r(m_2, w)$  replaced by  $K_0 e^{-ar}$ , with the absolute constants from Theorem 10.*

*Proof. Step 1: fix the circuit stages and the event bound. Write*

$$C = L_4 D_4 L_3 D_3 C^{(2)}, \quad C^{(2)} = L_2 D_2 L_1 D_1 L_0,$$

where the  $L_j$  are products of one-qubit unitaries and the  $D_j$  are disjoint controlled- $Z$  layers. Let  $\rho_x^{(2)}$  be the state after  $C^{(2)}$ , and let  $\rho_x$  be the state after  $L_3 D_3 C^{(2)}$ , immediately before  $D_4$ . Absorb  $L_4$  into the final output measurement, denoting its one-qubit Hermitian unitary by  $A_o$ , and set

$$\alpha(B) = \mathbb{E}_{x \in \{0,1\}^r} \chi(x) \text{Tr}(\rho_x B), \quad \chi(x) = (-1)^{x_1 + \dots + x_r},$$

with a uniform average.

Fix  $w \geq 1$ . In this proof,  $\beta$  denotes either  $\Delta_r(m_2, w)$  or  $K_0 e^{-ar}$ . Theorems 5 and 10, respectively, bound the absolute full-parity coefficient of every admissible two-layer port-product event by  $\beta$ . Product initialization and first-layer disjointness give  $r$  independent one-input sources. Group all input-free first-layer components into a single resource source; its state is a density operator tensor-independent of those sources. The resource ports are defined *after* this grouping, as in those theorems. Thus the density-resource hypotheses of both bounds are satisfied.

If  $D_4$  does not touch  $o$ , Lemma 13 gives  $\alpha(A_o) = 1$ , and Lemma 12 with  $s = 1$  gives  $1 \leq 36\beta$ . This proves both assertions with  $N_{\text{cell}} = 0$ . Henceforth let  $G$  be the gate of  $D_4$  incident to  $o$ . All other gates in  $D_4$  cancel from its output expectation, as in Lemma 13. Let  $\mathcal{T}$  be the set of control wires of  $G$ , excluding  $o$ . Write

$$P = \bigotimes_{q \in \mathcal{T}} P_q, \quad Q = I - P, \quad G = Q \otimes I_o + P \otimes R_o,$$

where  $P_q$  are one-qubit rank-one orthogonal projectors and  $R_o$  is the induced one-qubit reflection.

*Step 2: partition the controls into cells.* Let  $S_1, \dots, S_r$  be the input-containing first-layer source wire sets, and let  $S_0$  be the union of the input-free components;  $S_0$  may be empty. For  $\ell = 2, 3$ , let

$V_{\ell,j}$  be the support of gate  $j$  in layer  $\ell$ , for  $1 \leq j \leq m_\ell$ , and let  $V_{\ell,0}$  be the set of wires idle in that layer. Define the cells to be the nonempty sets

$$c_{ijk} = \mathcal{T} \cap S_i \cap V_{2,j} \cap V_{3,k}, \quad 0 \leq i \leq r, \quad 0 \leq j \leq m_2, \quad 0 \leq k \leq m_3.$$

Each wire has exactly one source label and one gate-or-idle label in each layer, so these cells partition  $\mathcal{T}$ . Their number satisfies

$$N_{\text{cell}} \leq (r+1)(m_2+1)(m_3+1).$$

For  $j > 0$ , the cell lies in the port  $S_i \cap V_{2,j}$ . For  $j = 0$ , it lies in the single private idle port  $S_i \cap V_{2,0}$ . The global idle label therefore does not merge different source ports. Likewise, the cell lies entirely in one third-layer gate if  $k > 0$ , and is entirely idle if  $k = 0$ . The output is excluded from every cell, even if it has the same three labels. Tensor-factor reorderings used below are not physical SWAP gates.

*Step 3: apply the exact identity-sector argument to cells.* For a cell  $c$ , put

$$P_c = \bigotimes_{q \in c} P_q, \quad Q_c = I_c - P_c,$$

with identity extensions off  $c$ . The cells partition the controls, so  $P = \prod_c P_c$  and  $Q \leq \sum_c Q_c$ . Each  $Q_c$  commutes with  $A_o$  and  $G$ , annihilates  $P$ , and satisfies  $Q_c G = G Q_c = Q_c$ . The positive-compression argument in Lemma 13 therefore applies to  $Q_c$ , regardless of the dimension of the cell. It gives

$$\alpha(A_o Q_c) = \mathbb{E}_x \text{Tr}(\rho_x Q_c) \geq 0.$$

Combining the operator union bound with the comparison  $GA_oG - R_oA_oR_o = Q(A_o - R_oA_oR_o)$ , exactly as in that lemma, yields

$$1 \leq |\alpha(R_oA_oR_o)| + 2 \sum_c |\alpha(A_o Q_c)|. \quad (26)$$

This uses neither independence of the cells nor commutation of  $\rho_x$  with their projectors. It is a reuse of the proof's identity-sector argument, not an assertion that a cell is one physical qubit.

*Step 4: pull back one whole-cell projector.* We prove, for each fixed cell  $c$ ,

$$|\alpha(A_o P_c)| \leq 1296\beta.$$

Write  $L_{3,q}$  for the one-qubit factor of  $L_3$  on wire  $q$ , and put  $L_{3,c} = \bigotimes_{q \in c} L_{3,q}$ . Define

$$\tilde{A}_o = L_{3,o}^* A_o L_{3,o}, \quad \tilde{P}_c = L_{3,c}^* P_c L_{3,c}.$$

Then  $L_3^*(A_o P_c)L_3 = \tilde{A}_o \tilde{P}_c$ , where  $\tilde{A}_o$  is a one-qubit contraction and  $\tilde{P}_c$  is still a product of one-qubit rank-one orthogonal projectors on  $c$ . Cyclicity of trace gives

$$\text{Tr}(\rho_x A_o P_c) = \text{Tr} \left[ \rho_x^{(2)} D_3 (\tilde{A}_o \tilde{P}_c) D_3 \right].$$

The observable meets at most two gates in  $D_3$ : one at  $o$  and one containing  $c$ . These gates may coincide, or either may be absent. All other third-layer gates conjugate identity.

For an affected gate  $D_g = I - 2\Pi_g$ , expand its conjugation as

$$D_g B D_g = B - 2\Pi_g B - 2B\Pi_g + 4\Pi_g B\Pi_g.$$

Keep the coefficients  $1, -2, -2, 4$  separate from the operator factors. If  $h \leq 2$  gates are affected, their simultaneous expansion has coefficient absolute sum at most  $9^h \leq 81$ . In each scalar-free branch, the output factor is a one-qubit contraction. If  $c$  is in gate  $g$ , let  $\Pi_c$  be the tensor product of the trigger projectors of  $\Pi_g$  on  $c$ . The cell factor is one of

$$\tilde{P}_c, \quad \Pi_c \tilde{P}_c, \quad \tilde{P}_c \Pi_c, \quad \Pi_c \tilde{P}_c \Pi_c.$$

Each has rank at most one and norm at most one. If  $c$  is idle, its factor is just  $\tilde{P}_c$ . Every physical-qubit factor outside  $\{o\} \cup c$  is identity or a rank-one orthogonal projector. When the output and cell meet the same gate, the branch choices on them are synchronized, but each branch still factors over the disjoint blocks  $\{o\}$ ,  $c$ , and the remaining physical wires.

For completeness, let  $M$  be a rank-at-most-one contraction on an arbitrary finite-dimensional space. If  $M = 0$ , omit the term. Otherwise write  $M = uv^*$  and put

$$H = (M + M^*)/2, \quad K = (M - M^*)/(2i).$$

Both are Hermitian contractions whose ranges are contained in  $\text{span}\{u, v\}$ . Hence each has at most two nonzero eigenvalues, all of absolute value at most one. Their separate spectral decompositions express  $M = H + iK$  as a linear combination of rank-one orthogonal projectors with coefficient absolute sum at most four. Repeated eigenvalues admit an orthonormal eigenbasis; zero eigenvalues contribute nothing. The coefficients need not be positive or real. The one-qubit output contraction also admits coefficient mass at most four, by the decomposition in Lemma 12.

Applying these decompositions separately on the output and cell blocks gives a finite, input-independent expansion

$$D_3(\tilde{A}_o \tilde{P}_c) D_3 = \sum_{\nu} d_{\nu} E_{\nu}, \quad \sum_{\nu} |d_{\nu}| \leq 81 \cdot 4 \cdot 4 = 1296.$$

A spectral projector on  $c$  may be entangled across its physical wires. Nevertheless,  $c$  lies in one source-second-layer port. Regrouping every  $E_{\nu}$  by those ports therefore gives a tensor product of positive contractions. If the output or other affected wires share the cell's port, their positive factors act on disjoint tensor factors within that port. Their tensor product with the cell projector remains a positive contraction. This also applies to ports on the grouped resource source.

To apply the two-layer bound, let  $\tau_x$  be the state immediately before  $D_2$ , so  $\rho_x^{(2)} = L_2 D_2 \tau_x D_2 L_2^*$ . Set  $F_{\nu} = L_2^* E_{\nu} L_2$ . Local conjugation preserves positivity and port factorization, including any entangled cell projector, and

$$\text{Tr}(\rho_x^{(2)} E_{\nu}) = \text{Tr}(F_{\nu} D_2 \tau_x D_2).$$

The event theorem applied to  $F_{\nu}$  therefore gives

$$\left| \mathbb{E}_x \chi(x) \text{Tr}(\rho_x^{(2)} E_{\nu}) \right| \leq \beta.$$

Linearity and the triangle inequality prove  $|\alpha(A_o P_c)| \leq 1296\beta$ . No physical-product-only event bound has been substituted for the port-product bound in this step.

*Step 5: sum the cell certificate.* Lemma 12 with  $s = 1$  gives

$$|\alpha(A_o)|, |\alpha(R_o A_o R_o)| \leq 36\beta.$$

Since  $\alpha(A_o Q_c) = \alpha(A_o) - \alpha(A_o P_c)$ ,

$$|\alpha(A_o Q_c)| \leq (36 + 1296)\beta = 1332\beta.$$

Substitution into (26) yields

$$1 \leq 36\beta + 2N_{\text{cell}}(1332\beta) = [36 + 2664N_{\text{cell}}]\beta.$$

The cell partition is independent of  $w$  and of the choice of  $\beta$ , so this proves both versions of (25) with the same  $N_{\text{cell}}$ .  $\square$

*Proof of Theorem 1.* Let  $C$  be an exact  $n$ -input circuit as in the theorem, with  $M$  multiqubit gates. Apply Lemma 11, obtaining a product-initialized circuit on the same physical wires with  $\lceil n/3 \rceil \leq r \leq n$  variable inputs and at most  $M$  multiqubit gates. If its task is the complement of parity, reverse the output labels, replacing  $A_o$  by  $-A_o$ . This changes neither depth nor gate count. Pad with empty layers at the end if necessary to obtain four designated multiqubit layers. The first-layer source condition is preserved, and the second- and third-layer gate counts satisfy  $m_2, m_3 \leq M$ .

The exponential cell certificate gives

$$1 \leq [36 + 2664(r+1)(M+1)^2]K_0e^{-ar} \leq (M+1)^2K_0[36 + 2664(r+1)]e^{-ar}.$$

The second inequality uses  $(M+1)^2 \geq 1$ . All quantities are positive, so taking natural logarithms yields

$$\begin{aligned} 2\log(M+1) &\geq ar - \log(K_0[36 + 2664(r+1)]) \\ &\geq \frac{an}{3} - \log(K_0[36 + 2664(n+1)]). \end{aligned}$$

The last bound is uniform over all permitted values of  $r$ .

Here is an explicit absolute input threshold. Put  $C_0 = 2700K_0$  and choose

$$n_0 = \left\lceil \max \left\{ 1, \frac{12\log C_0}{a}, \frac{144}{a^2}, \frac{24\log 2}{a} \right\} \right\rceil.$$

For  $n \geq 1$ ,  $36 + 2664(n+1) \leq 2700(n+1)$  and  $\log(n+1) \leq \sqrt{n}$ . The latter follows since  $\sqrt{t} - \log(1+t)$  vanishes at  $t = 0$  and has derivative  $(\sqrt{t} - 1)^2/[2\sqrt{t}(1+t)] \geq 0$  for  $t > 0$ . Consequently, when  $n \geq n_0$ ,

$$\log(K_0[36 + 2664(n+1)]) \leq \log C_0 + \sqrt{n} \leq \frac{an}{6}.$$

It follows that  $2\log(M+1) \geq an/6$ , hence  $M+1 \geq e^{an/12}$ . Finally,  $t = e^{an/24} \geq 2$  for  $n \geq n_0$ , so

$$M \geq e^{an/12} - 1 = t^2 - 1 \geq t = e^{an/24}.$$

Thus Theorem 1 holds with  $c_* = a/24$  and the displayed  $n_0$ . These constants depend only on the absolute constants  $a, K_0$ , not on physical wire count or gate arities. No bound on the number of ancillary wires was used.  $\square$

## 10 Arbitrary first layers and a bounded-error final-arity bound

This section derives a size-independent exponential estimate for physical product events with no restriction on first-layer input arities. It then gives a bounded-error depth-four consequence when the final output gate has subquadratic arity. The latter is not a bounded-error lower bound for arbitrary polynomial final arity.

Throughout this section the circuit model, uniform input averages, and identity extensions of operators are those of Section 2. In particular,  $N \geq n \geq 1$ , initialization is pure and product on

the ancillary wires, and empty multiqubit layers and arbitrary boundary and intervening one-qubit gates are allowed. No exact-computation assumption is made here.

Fix the absolute constants  $K_0 \geq 1$  and  $a > 0$  from Theorem 10, so that its cleaned two-layer event estimate is  $K_0 e^{-ar}$  for  $r \geq 0$  one-input sources, including  $r = 0$ . The independent input-free resource in that theorem may have arbitrary finite dimension and, as stated there, may be an arbitrary operator of trace norm at most one. The latter extension is essential below because exact gate expansions need not leave a positive resource operator.

**Theorem 15** (Physical product events after arbitrary first layers). *There is an absolute constant  $\kappa > 0$  such that, after any two-layer prefix in the model above, every fixed, input-independent physical-qubit product event  $E$  satisfies*

$$|\mathbb{E}_x \chi(x) \langle E \rangle_x| \leq K_0 e^{-\kappa n}. \quad (27)$$

*The number of gates and ancillary wires is unrestricted and finite. The same statement holds for a physical tensor product of positive one-qubit contractions.*

*Proof. Step 1: circuit stages and the signed first-layer expansion.* Let  $q(x) = \langle E \rangle_x$  for the original circuit. Write  $C^{(2)} = L_2 D_2 L_1 D_1 L_0$ , with local layers  $L_j$  and disjoint reflection layers  $D_j$  as in Section 2. Put  $E' = L_2^* E L_2$  and use  $E'$  as the event immediately after  $D_2$  in the trace functionals below. This leaves  $q(x)$  unchanged and preserves the physical-qubit product form. The local layer  $L_1$  conjugates each first-layer component separately, so it preserves its trace norm, independence from other components, and any splitting into individual wires used below.

Fix an integer  $L \geq 1$ , to be chosen as an absolute constant below. Call a first-layer gate *large* if it contains  $k > L$  original input wires. On its support let  $\tau_x$  be the product input state immediately after  $L_0$ , with  $x \in \{0, 1\}^k$  denoting these inputs. Its normalized signed operator is

$$F_0 = \mathbb{E}_x (-1)^{x_1 + \dots + x_k} \tau_x = \bigotimes_{j=1}^k \frac{S_j}{2} \otimes \bigotimes_{q \in \text{anc}} |\alpha_q\rangle \langle \alpha_q|,$$

where  $S_j = \rho_{j,0} - \rho_{j,1}$  is the difference of the two orthogonal one-qubit input states after  $L_0$ , and  $\alpha_q$  here denotes the fixed ancillary unit vector after  $L_0$ . Thus every  $S_j$  is a traceless Hermitian unitary, and the factor  $1/2$  comes from the signed uniform average over that input bit. This formula includes arbitrary initial local input bases and also shows  $\|F_0\|_1 = 1$ .

For  $D = I - 2\Pi$ , expand the signed operator exactly:

$$D F_0 D = F_0 - 2\Pi F_0 - 2F_0 \Pi + 4\Pi F_0 \Pi. \quad (28)$$

The baseline  $F_0$  is the signed average of the same product input family with this first-layer gate removed. Thus its input variables can be kept as singleton sources and averaged later. This is an exact linear identity, not a postselection or a claim that a signed operator is a state. Each of the other three raw operators is still a tensor product on physical wires. On an input wire, the relevant factors are

$$\Pi_j S_j / 2, \quad S_j \Pi_j / 2, \quad \Pi_j S_j \Pi_j / 2,$$

each with trace norm at most  $1/2$ . For the first two this follows from unitarity of  $S_j$  and rank one of  $\Pi_j$ ; for the third it follows from  $|\text{Tr}(\Pi_j S_j)| \leq 1$ . On an ancillary wire the corresponding factor has trace norm at most one. Multiplicativity therefore gives

$$\|\Pi F_0\|_1, \|F_0 \Pi\|_1, \|\Pi F_0 \Pi\|_1 \leq 2^{-k}. \quad (29)$$

The correction coefficients in (28) have absolute sum  $2 + 2 + 4 = 8$ . For each nonzero raw correction, extract a factor  $2^{-k}$ ; its remaining operator has trace norm at most one and need not be Hermitian or positive. The  $k$  original input variables of that component have already been integrated with their parity signs in  $F_0$ . Thus the remaining operator is an input-free resource, independent of all still-variable components. Zero corrections may simply be discarded.

*Step 2: the resource and variables in a fixed branch.* Let  $\mathcal{H}$  be the set of large first-layer gates. For a subset  $S \subseteq \mathcal{H}$ , select one of the three raw corrections at each gate in  $S$ , recording these choices by  $\eta \in \{1, 2, 3\}^S$ , and take the baseline at the other large gates. Put

$$K(S) = \sum_{g \in S} k_g.$$

The selected components are disjoint. Their combined raw norm costs at most  $2^{-K(S)}$ , and the sum of absolute scalar coefficients over their three correction choices is  $8^{|S|}$ . After extracting the norm cost, tensor their operators with all other input-free resource factors. The resulting independent resource has trace norm at most one and is admissible in Theorem 10. Here the other input-free factors include the density operators produced by original input-free first-layer gates. They can be internally entangled. They are tensor-independent of the selected components and of all remaining variable sources. The whole resource is grouped before its source–second-layer ports are defined.

There remain  $n - K(S)$  variable inputs. An unselected large gate has become its baseline, so its input wires are separate singleton sources and its ancillary wires are fixed resource wires. Every unexpanded small first-layer component contains at most  $L$  inputs. Keep one chosen variable from each remaining input-bearing small component, keep all baseline and originally idle singleton variables, and fix the other remaining variables to computational values. If  $r$  variables were kept, then

$$r \geq \frac{n - K(S)}{L}. \quad (30)$$

The set of kept variables depends only on  $S$  and the circuit, not on the values fixed. Each conditioned component is a one-input source with fixed product ancillas, including the fixed original inputs in arbitrary local bases. Thus it satisfies the hypotheses of Theorem 10.

Here the physical-product-event qualification is essential. The original event factors on individual wires, so after any baseline splitting it still factors over every new source–second-gate port. An event that factors only over the old source ports could be entangled across the new source boundaries and need not remain admissible. No such broader event class is asserted here.

*Step 3: condition, apply the source theorem, and average.* For the chosen  $(S, \eta)$ , let  $q_{S, \eta}(y, z)$  be the trace functional after the second layer with the extracted factor  $2^{-K(S)}$  and the coefficients  $-2, -2, 4$  omitted. Here  $y$  are the kept variables and  $z$  the other remaining variables. Its resource has trace norm at most one; consequently this functional need not be a probability or even be real. For every fixed assignment of  $z$ , Theorem 10 bounds the coefficient on the kept variables by  $K_0 e^{-ar}$ . If  $B_{S, \eta}$  is the scalar branch before extracting  $2^{-K(S)}$  but still omitting the coefficients  $-2, -2, 4$ , then

$$B_{S, \eta} = 2^{-K(S)} \mathbb{E}_z \chi(z) [\mathbb{E}_y \chi(y) q_{S, \eta}(y, z)].$$

The characters here are parity on their respective variable sets. An empty parity character has value one, and an average over an empty variable set evaluates its unique summand. All averages are normalized, so no factor exponential in the number of fixed variables appears. Triangle inequality and (30) consequently bound each raw branch associated with  $S$  by

$$|B_{S, \eta}| \leq K_0 2^{-K(S)} e^{-a(n - K(S))/L}. \quad (31)$$

This includes  $K(S) = n$ : there are then no retained variables, and the norm-one resource bound is covered by  $K_0 \geq 1$ .

*Step 4: sum the branches and choose an absolute cutoff.* Summing (31) over all subsets and correction types gives the exact expansion estimate

$$\begin{aligned} |\hat{q}([n])| &\leq K_0 e^{-an/L} \sum_{S \subseteq \mathcal{H}} 8^{|S|} e^{-(\log 2 - a/L)K(S)} \\ &= K_0 e^{-an/L} \prod_{g \in \mathcal{H}} \left(1 + 8e^{-k_g(\log 2 - a/L)}\right). \end{aligned} \quad (32)$$

Choose a fixed integer  $L$  sufficiently large that

$$\log 2 - a/L > 0, \quad 8e^{-(L+1)(\log 2 - a/L)} \leq a/2. \quad (33)$$

Such an integer exists: the second left-hand side tends to zero as  $L \rightarrow \infty$ , while  $a$  is a fixed positive constant. First-layer disjointness gives  $|\mathcal{H}| \leq n/(L+1) \leq n/L$ . Since  $\log(1+t) \leq t$  for  $t \geq 0$ , the product in (32) has logarithm at most

$$\frac{n}{L} 8e^{-(L+1)(\log 2 - a/L)} \leq \frac{an}{2L}.$$

Equation (27) follows with the absolute constant  $\kappa = a/(2L)$ . No additive gate-deletion error was introduced, so the cutoff  $L$  does not need to grow with  $n$ . The argument does not count input-free gates or ancillary wires.

If the event has arbitrary positive one-qubit contraction factors, the same source-port factorization and cleaned theorem apply directly. Alternatively, if its eigenvalues are  $0 \leq \lambda_1 \leq \lambda_2 \leq 1$ , the factor equals  $\lambda_1 I + (\lambda_2 - \lambda_1)P + (1 - \lambda_2)0$  for a rank-one projector  $P$  onto an eigenvector with eigenvalue  $\lambda_2$ . If the eigenvalues coincide, any rank-one projector may be used. Thus it is a convex combination of identity, a rank-one projector, and zero. Expanding these convex combinations proves the same estimate.  $\square$

**Corollary 16** (Small-support observables with arbitrary first layers). *Consider any three-layer prefix in the model above. Let  $s \geq 0$  be an integer, let  $S$  be a fixed set of at most  $s$  physical wires, and let  $B = (\bigotimes_{q \in S} B_q) \otimes I_{S^c}$ , where each  $B_q$  is an input-independent one-qubit contraction. At the output of this prefix,*

$$|\mathbb{E}_x \chi(x) \langle B \rangle_x| \leq 36^s K_0 e^{-\kappa n}. \quad (34)$$

*Hermiticity is not required, and the coefficient may be complex. The constants are those of Theorem 15.*

*Proof.* Use the algebraic pullback expansion in the proof of Lemma 12; that expansion does not assume a cleaned first layer. Put  $t = |S| \leq s$ . After absorbing the final local layer into  $B$ , at most  $h \leq t$  third-layer gates meet its support. Their reflection expansions have coefficient mass at most  $9^h$ . Off the original  $t$  sites, each factor is identity or a rank-one projector. Polarizing the remaining one-qubit contractions costs at most  $4^t$ , because each is the sum of a Hermitian contraction and  $i$  times another Hermitian contraction, and each has a two-eigenvector spectral decomposition. The resulting terms are input-independent physical-qubit product events, with coefficient mass at most  $9^h 4^t \leq 36^s$ . The local layer after layer two is absorbed into these events without changing their form. Apply Theorem 15, rather than a cleaned-first-layer estimate, to every expanded term. This proves (34). For  $S = \emptyset$ , the observable is identity and its coefficient is zero because  $n \geq 1$ . The argument also allows  $s > N$ , since only the actual support size  $t$  is used.  $\square$

**Corollary 17** (Bounded error with subquadratic final arity). *Let  $C$  be a four-layer circuit in the model above with a binary one-qubit output measurement, and let  $c = c(C, A_o)$  be its uniform-input parity correlation from (3). If a gate in the fourth multiqubit layer contains the measured output, let  $b \geq 1$  be its number of controls, so that its total arity is  $b + 1$ . If no such gate exists, put  $b = 0$ . For  $b \geq 2$  and every integer  $\ell \geq 0$ ,*

$$|c| \leq K_0(36 + 72 \cdot 540^\ell)e^{-\kappa n} + 2 \operatorname{sech}\left(2\ell \operatorname{artanh} \frac{1}{\sqrt{b}}\right). \quad (35)$$

*For  $b = 0$  and  $b = 1$  the respective bounds are  $|c| \leq 36K_0e^{-\kappa n}$  and  $|c| \leq 2628K_0e^{-\kappa n}$ . In particular,  $b = o(n^2)$  implies  $|c| = o(1)$ , independently of the number of gates and product ancillary wires. Hence such families cannot compute parity with any fixed positive average or worst-case success advantage.*

*Proof. Step 1: the final observable and its control-count operator.* Discard fourth-layer gates disjoint from the output. Absorb its final local rotation into the Hermitian contraction  $A_o$  from (3). This also covers a general binary one-qubit measurement. In the remainder of this proof  $\rho_x$  and  $\langle \cdot \rangle_x$  refer to the state immediately before the fourth multiqubit layer, not to the full output state in (3). If  $b = 0$ , then  $c = \mathbb{E}_x \chi(x) \langle A_o \rangle_x$  and Corollary 16 with  $s = 1$  proves the stated bound. Otherwise let  $G$  be the unique output-incident gate. In the final reflection basis write

$$G = (I - P) \otimes I_o + P \otimes R_o, \quad P = \prod_{j=1}^b P_j, \quad A_\perp = \frac{A_o - R_o A_o R_o}{2},$$

where  $P_j$  is the fixed rank-one orthogonal trigger projector on control wire  $j$ , these wires exclude  $o$ , and  $R_o$  is the one-qubit reflection induced by the final gate on the output. The operators  $P$  and  $A_o$  act on disjoint registers. Thus the cross terms in  $GA_oG$  vanish by  $P(I - P) = 0$ , giving  $GA_oG = (I - P)A_o + PR_oA_oR_o$ . Moreover,  $\|A_\perp\|_\infty \leq (\|A_o\|_\infty + \|R_oA_oR_o\|_\infty)/2 \leq 1$ , so  $A_\perp$  is a one-qubit Hermitian contraction, and

$$GA_oG = A_o - 2PA_\perp. \quad (36)$$

No commutation between  $A_o$  and  $R_o$  is assumed. For  $b = 1$ , the two terms in this identity are supported on at most one and two wires, respectively. Applying Corollary 16 gives  $(36 + 2 \cdot 36^2)K_0e^{-\kappa n} = 2628K_0e^{-\kappa n}$ . Henceforth assume  $b \geq 2$ . Define commuting projections  $Q_j = I - P_j$ , and put  $K = \sum_{j=1}^b Q_j$  and  $J = K/b$ . The spectrum of  $K$  is contained in  $\{0, 1, \dots, b\}$ , and  $P$  is its zero-eigenspace projector.

*Step 2: approximate the zero-eigenspace projector.* Let  $T_\ell$  be the Chebyshev polynomial of the first kind, defined by  $T_0(z) = 1$ ,  $T_1(z) = z$ , and  $T_{\ell+1}(z) = 2zT_\ell(z) - T_{\ell-1}(z)$  for  $\ell \geq 1$ . We use its standard identities  $T_\ell(\cos \theta) = \cos(\ell\theta)$  and  $T_\ell(\cosh t) = \cosh(\ell t)$ ; see, for example, [18]. Set

$$p_\ell(K) = \frac{T_\ell((b+1-2K)/(b-1))}{T_\ell((b+1)/(b-1))}.$$

This polynomial is one at zero, and on the rest of the spectrum has absolute value at most

$$\nu_\ell = \frac{1}{T_\ell((b+1)/(b-1))} = \operatorname{sech}\left(2\ell \operatorname{artanh} \frac{1}{\sqrt{b}}\right) \leq 2e^{-2\ell/\sqrt{b}}. \quad (37)$$

In particular,  $\|p_\ell(K) - P\|_\infty \leq \nu_\ell$ . Indeed, on every eigenspace with eigenvalue  $1 \leq k \leq b$ , the numerator's argument lies in  $[-1, 1]$ , and  $\operatorname{arcosh}((b+1)/(b-1)) = 2 \operatorname{artanh}(1/\sqrt{b}) \geq 2/\sqrt{b}$ . The

last estimate in (37) also uses  $\text{sech } t \leq 2e^{-t}$  for  $t \geq 0$ . For  $\ell = 0$ ,  $p_0(K) = I$  and  $\nu_0 = 1$ ; the same formulas hold, although the approximation bound is then trivial.

*Step 3: control coefficient mass and physical support.* The coefficient mass is controlled in powers of the normalized variable  $J$ , not powers of  $K$ . Write the numerator's argument as  $u - vJ$ , where

$$u = \frac{b+1}{b-1}, \quad v = \frac{2b}{b-1}, \quad u + v \leq 7.$$

If  $N_\ell$  is the absolute coefficient sum of  $T_\ell(u - vJ)$ , its recurrence gives  $N_0 = 1$ ,  $N_1 \leq 7$ , and  $N_{\ell+1} \leq 14N_\ell + N_{\ell-1}$ . Induction using  $14 \cdot 15 + 1 \leq 15^2$  yields  $N_\ell \leq 15^\ell$ . The denominator defining  $p_\ell$  is at least one, so we can write  $p_\ell(K) = \sum_{k=0}^\ell a_k J^k$  with  $\sum_{k=0}^\ell |a_k| \leq 15^\ell$ .

For every  $k \geq 1$ ,

$$J^k = b^{-k} \sum_{j_1, \dots, j_k=1}^b Q_{j_1} \cdots Q_{j_k}.$$

This is a convex combination of products supported on at most  $k$  distinct controls; repeated indices are removed using  $Q_j^2 = Q_j$ . The  $k = 0$  term is identity. The output is not a control, so  $A_\perp$  acts on a separate wire. Consequently  $p_\ell(K)A_\perp$  is a linear combination of product observables on at most  $\ell + 1$  wires, with total coefficient mass at most  $15^\ell$ . Each such observable meets at most  $\ell + 1$  third-layer gates, regardless of their arities. Corollary 16 therefore gives

$$|\mathbb{E}_x \chi(x) \langle A_o - 2p_\ell(K)A_\perp \rangle_x| \leq K_0 \left( 36 + 2 \cdot 15^\ell 36^{\ell+1} \right) e^{-\kappa n}.$$

*Step 4: transfer the operator approximation to correlation.* Replacing  $p_\ell(K)$  by  $P$  changes the observable's operator norm by at most  $2\nu_\ell$ , by the operator-norm ideal inequality and  $\|A_\perp\|_\infty \leq 1$ . To justify the corresponding correlation bound, let  $\Omega = \mathbb{E}_x \chi(x) \rho_x$  be the signed state immediately before the fourth layer. Triangle inequality and normalization give  $\|\Omega\|_1 \leq \mathbb{E}_x \|\rho_x\|_1 = 1$ ; no independence of the controls or commutation of  $\rho_x$  with their projections is needed. Trace-norm duality therefore shows that replacing  $p_\ell(K)$  by  $P$  changes the correlation by at most  $2\nu_\ell$ . Equation (35) follows, since  $2 \cdot 15^\ell 36^{\ell+1} = 72 \cdot 540^\ell$ .

*Step 5: the subquadratic-arity consequence.* For an explicit vanishing bound, choose

$$\ell = \left\lfloor \frac{\kappa n}{2 \log 540} \right\rfloor.$$

The first term in (35) is at most  $108K_0 e^{-\kappa n/2}$ . If  $b = o(n^2)$  and  $b \geq 2$ , then  $\ell/\sqrt{b} \rightarrow \infty$ , so (37) shows that the other term also tends to zero. Together with the bounds already proved for  $b = 0, 1$ , this establishes the conclusion for any sequence  $b = b(n) = o(n^2)$ , including bounded or oscillating sequences. Neither the constants nor this choice of  $\ell$  depends on the total gate count or the ancillary-wire count.

Finally, success probability at least  $1/2 + \delta$  on uniform inputs implies  $c \geq 2\delta$ , for the chosen output labeling. A worst-case success guarantee implies the same average guarantee. Thus  $|c| = o(1)$  excludes every fixed  $\delta > 0$ , without amplification, extra input copies, or input preservation.  $\square$

*Remark 18 (An explicit fixed-advantage consequence).* For  $b \geq 2$  and fixed  $0 < \delta \leq 1/2$ , uniform-input success probability at least  $1/2 + \delta$  implies

$$\sqrt{b} \geq \frac{2[\kappa n - \log(108K_0/\delta) - \log 540] * +}{\log 540 \log(4/\delta)}, \quad [t] * + = \max\{t, 0\}.$$

This is a quadratic final-arity requirement at each fixed advantage, not a restriction on the total circuit size and not a lower bound for every polynomial choice of final arity.

*Proof.* Choose  $\ell = \lceil (\sqrt{b}/2) \log(4/\delta) \rceil$ . Equation (37) gives  $2\nu_\ell \leq 4e^{-2\ell/\sqrt{b}} \leq \delta$ . Since  $c \geq 2\delta$ , equation (35) and  $540^\ell \geq 1$  imply

$$\delta \leq K_0(36 + 72 \cdot 540^\ell)e^{-\kappa n} \leq 108K_0 540^\ell e^{-\kappa n}.$$

Taking natural logarithms and using  $\ell \leq (\sqrt{b}/2) \log(4/\delta) + 1$  yields

$$\kappa n - \log(108K_0/\delta) - \log 540 \leq \frac{\sqrt{b}}{2} \log 540 \log(4/\delta).$$

The denominator in the stated bound is positive. If the left-hand side is negative the bound is trivial; otherwise division gives the result. A worst-case success guarantee implies the uniform-input hypothesis and therefore the same conclusion.  $\square$

**Use of AI:** The author prepared an 8-page draft (without using AI) with outlines of previous QAC lower bounds and several high-level proof strategies to extend them to depths 4, 5 and unbounded. The author then fed this draft to multiple AI tools (Claude, ChatGPT and two open-weight models with open-source harnesses) to work out the details of the proofs and identify missing gaps. The author fixed those missing gaps and iteratively verified and expanded the proofs using AI tools. The author is solely responsible for the correctness of the proofs.

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